Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 21, 2026

1 Overview

This document compiles every current SCM case report TeX file under results/.

2 Case 1: sheba 160x2m 12h report

3 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

3.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 100 / 0
- RHS evaluations: 49002
- Solver tolerances: abstol = 1.000×10^{-8}, reltol = 1.000×10^{-6}

3.2 Forcing and Boundary Conditions

Table 1: Forcing and Boundary Condition Parameters

Parameter	**Symbol**	**Value**
Zonal geostrophic wind	U_g	$7.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	1.000e-04 m
Surface roughness (heat)	z_{0h}	1.000e-05 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	257.000000 K
Deep soil temperature	T_{deep}	255.500000 K
Downward longwave forcing	$R_{\downarrow}$	$190.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	257.000000 K
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

3.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.162000%
- Diffusivity bounds: $K_m \in [0.017833, 6.615549]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.020467, 723.889862]$

3.4 Generated Artifacts

- Time-series CSV: `results/sheba/time_series.csv`
- Diagnostic summary JSON: `results/sheba/summary.json`
- Payload JLD2: `results/sheba/payload.jld2`

3.5 Figure Manifest

1. Time series of T__s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R__n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e__xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

3.6 Included Plots

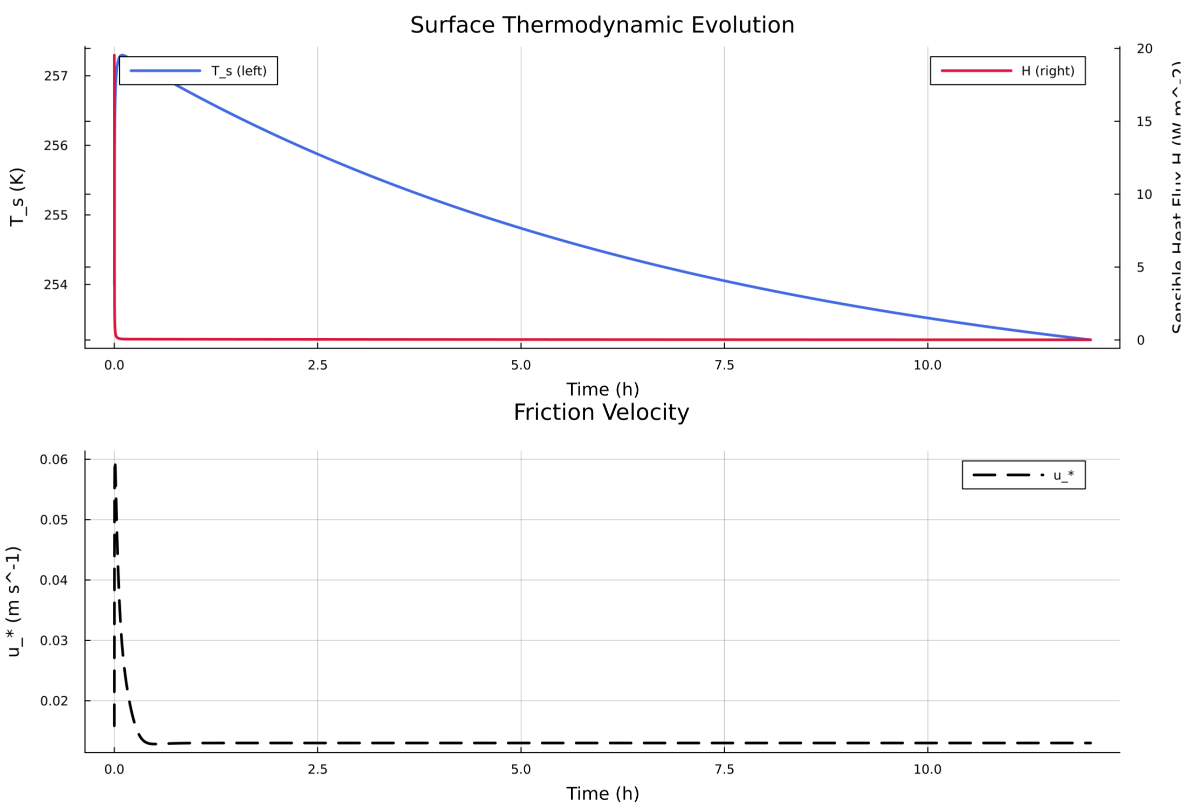

Figure 1: Time series of T__s, sensible heat flux, and friction velocity

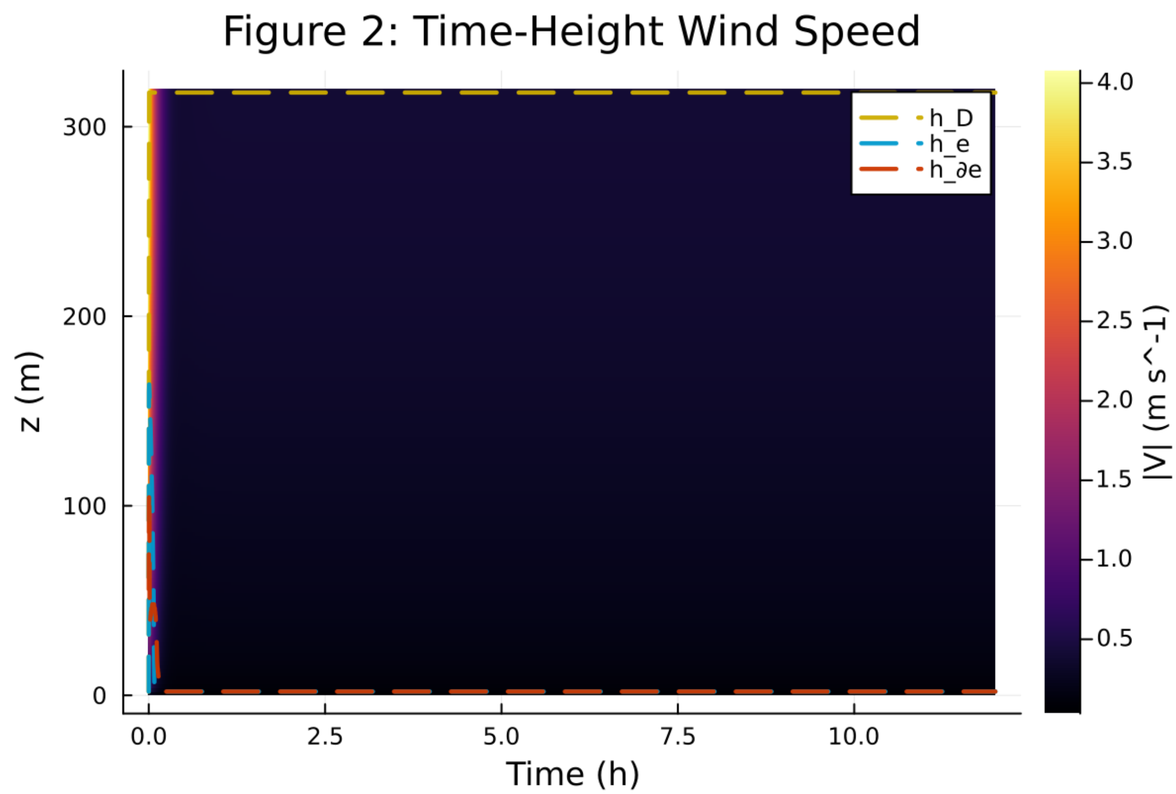

Figure 2: Time-height contour of wind speed with LLJ evolution

4 Case 2: gabls1 200x2m 12h report

5 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

5.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1441
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 99 / 0
- RHS evaluations: 60392
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

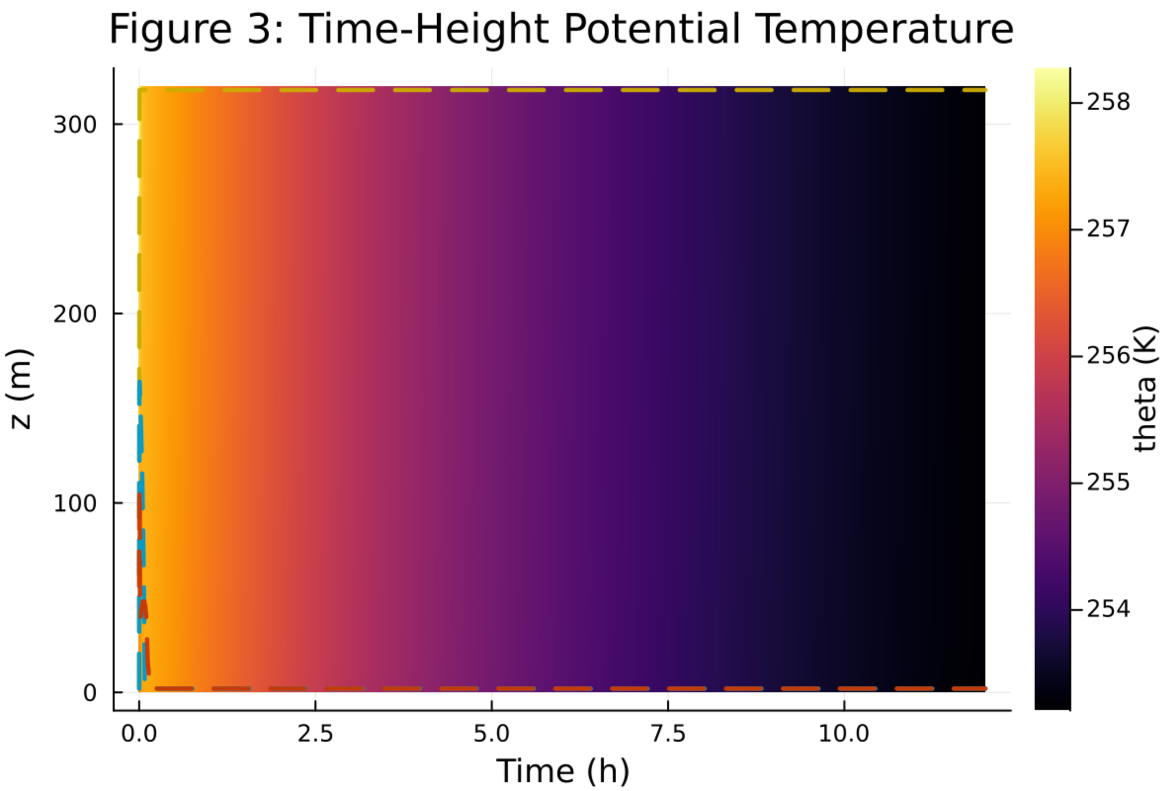

Figure 3: Time-height contour of potential temperature and inversion growth

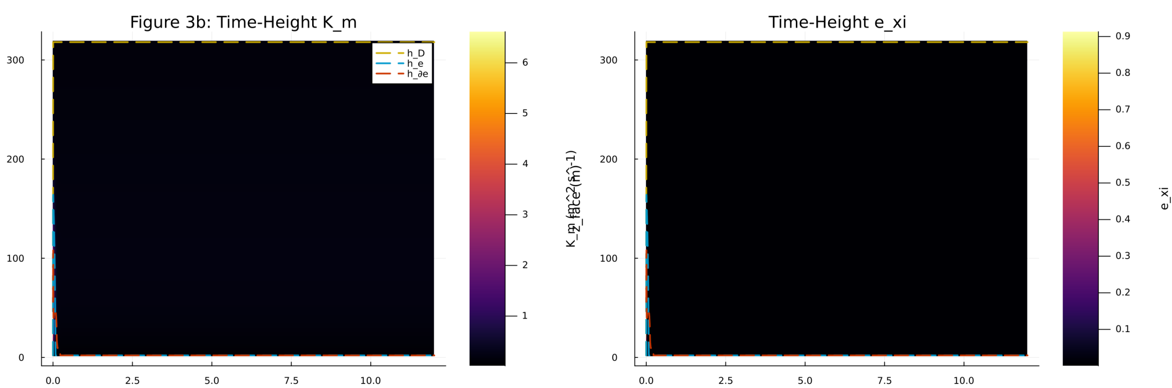

Figure 4: Vertical profiles of U, theta, and Km at representative times

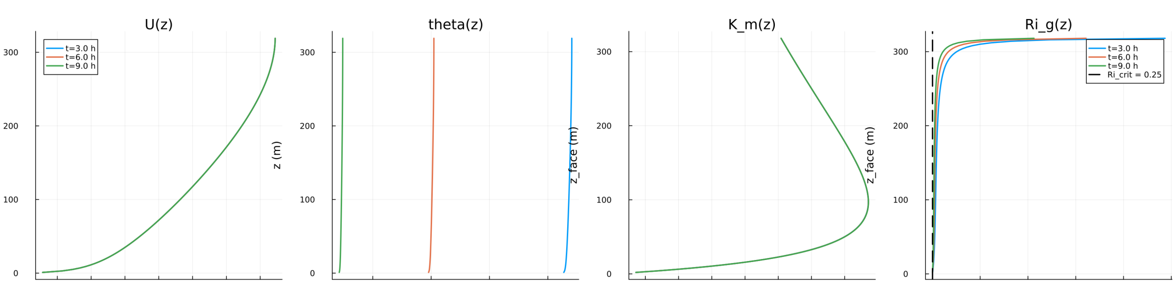

Figure 5: Surface energy budget components (R__n, H, G, storage)

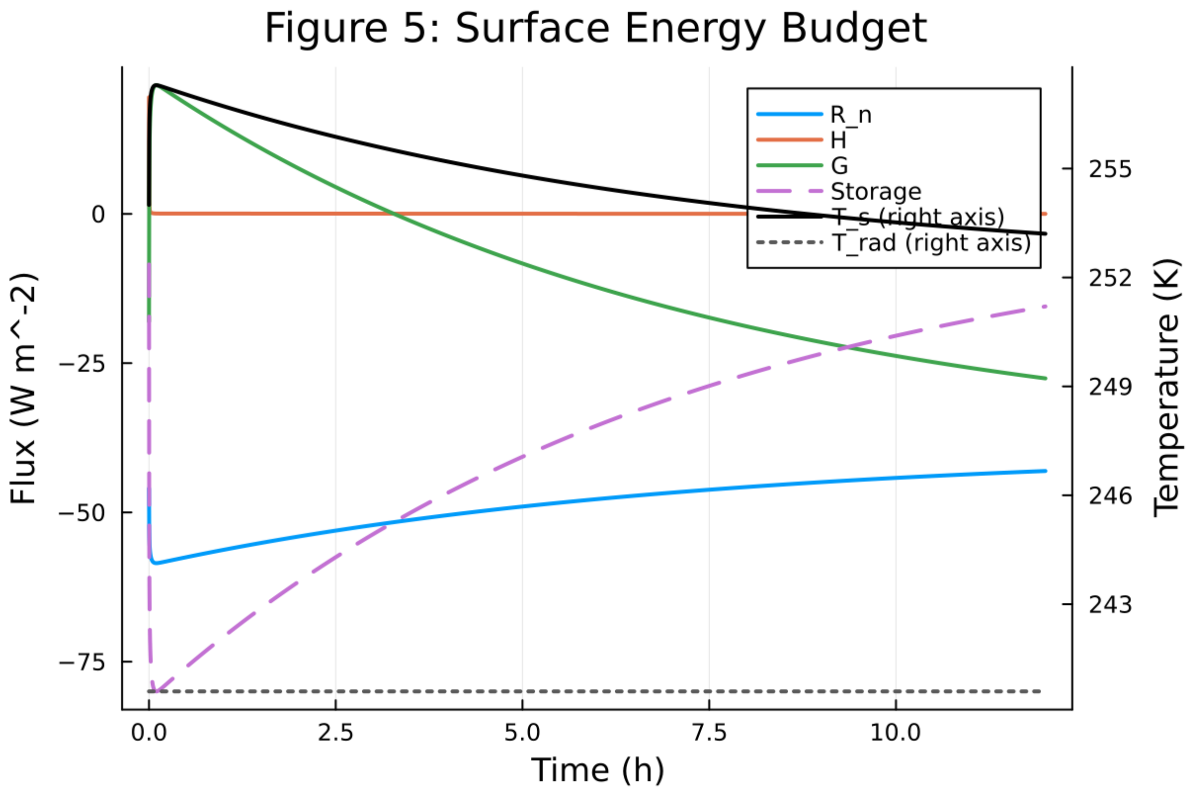

Figure 6: Phase portrait on regularized manifold (Delta vs e_xi)

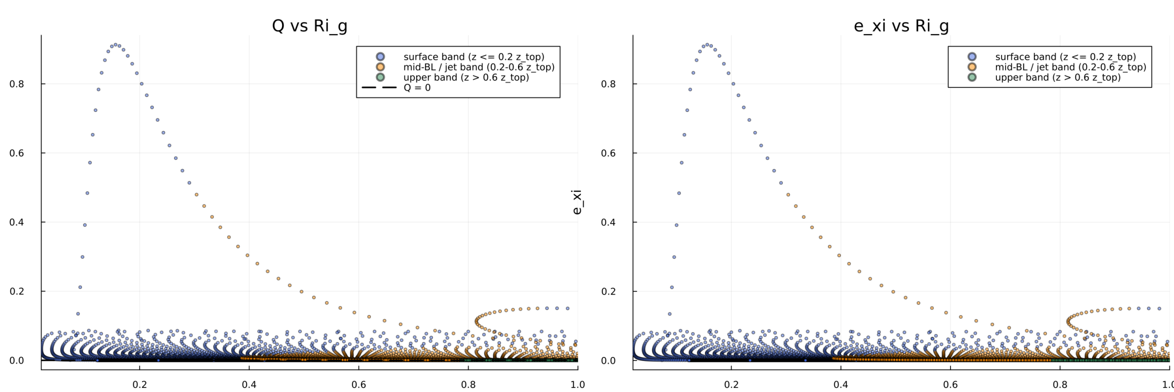

Figure 7: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

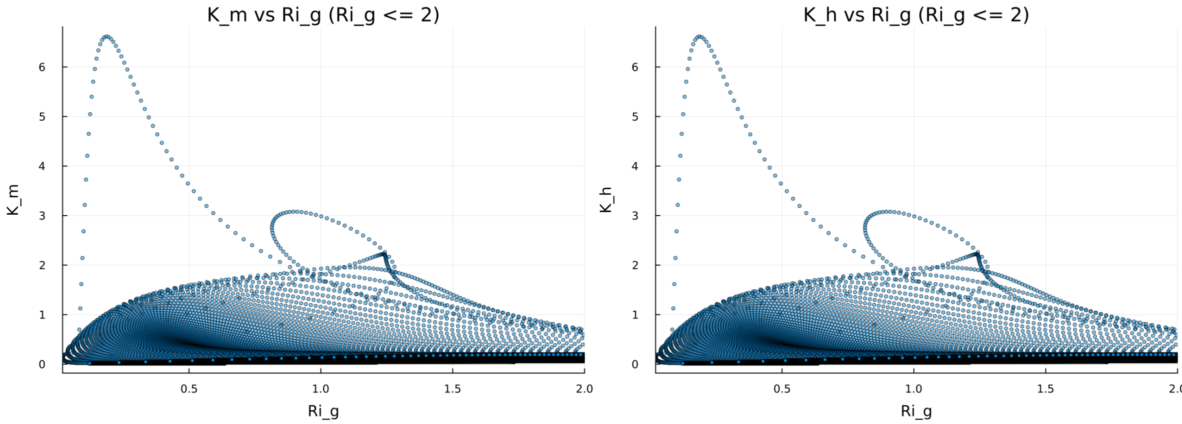

Figure 8: Fold proximity diagnostics versus time

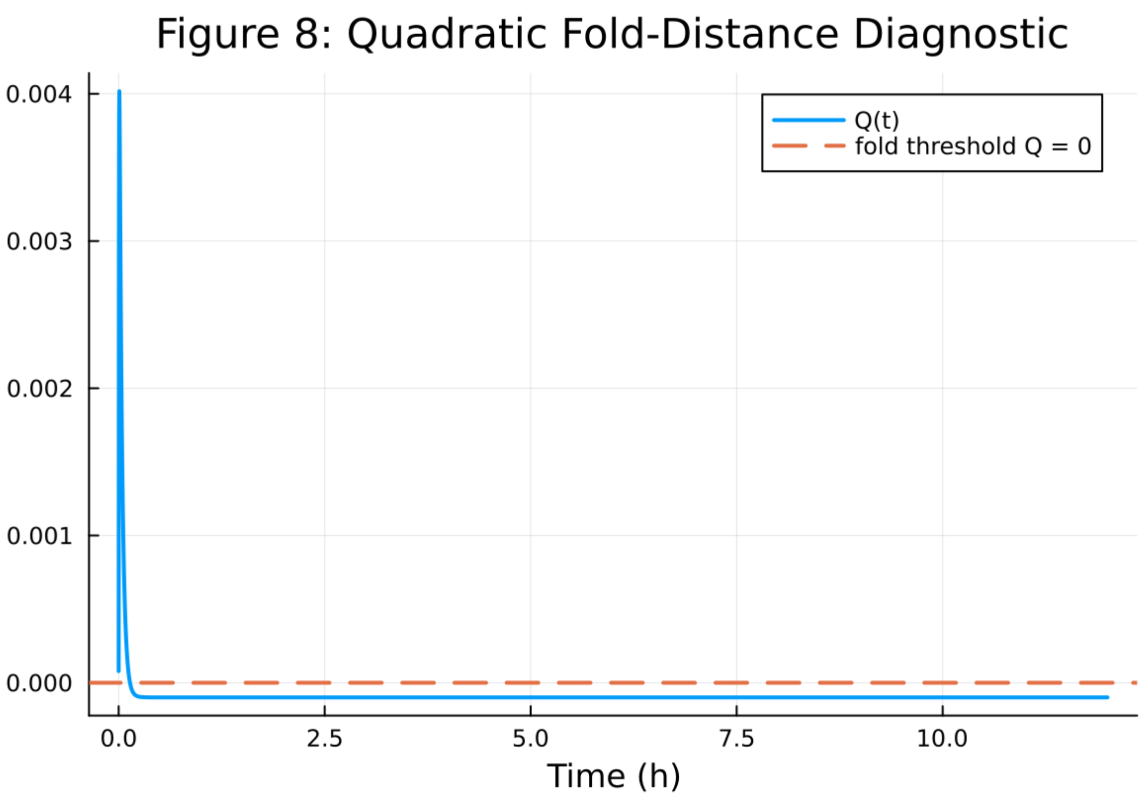

Figure 9: fig08 fold proximity.png

Table 2: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	8.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	245.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

5.2 Forcing and Boundary Conditions

5.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.346981%
- Diffusivity bounds: $K_m \in [0.006021, 3.564392]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.031448, 843.105664]$

5.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

5.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

5.6 Included Plots

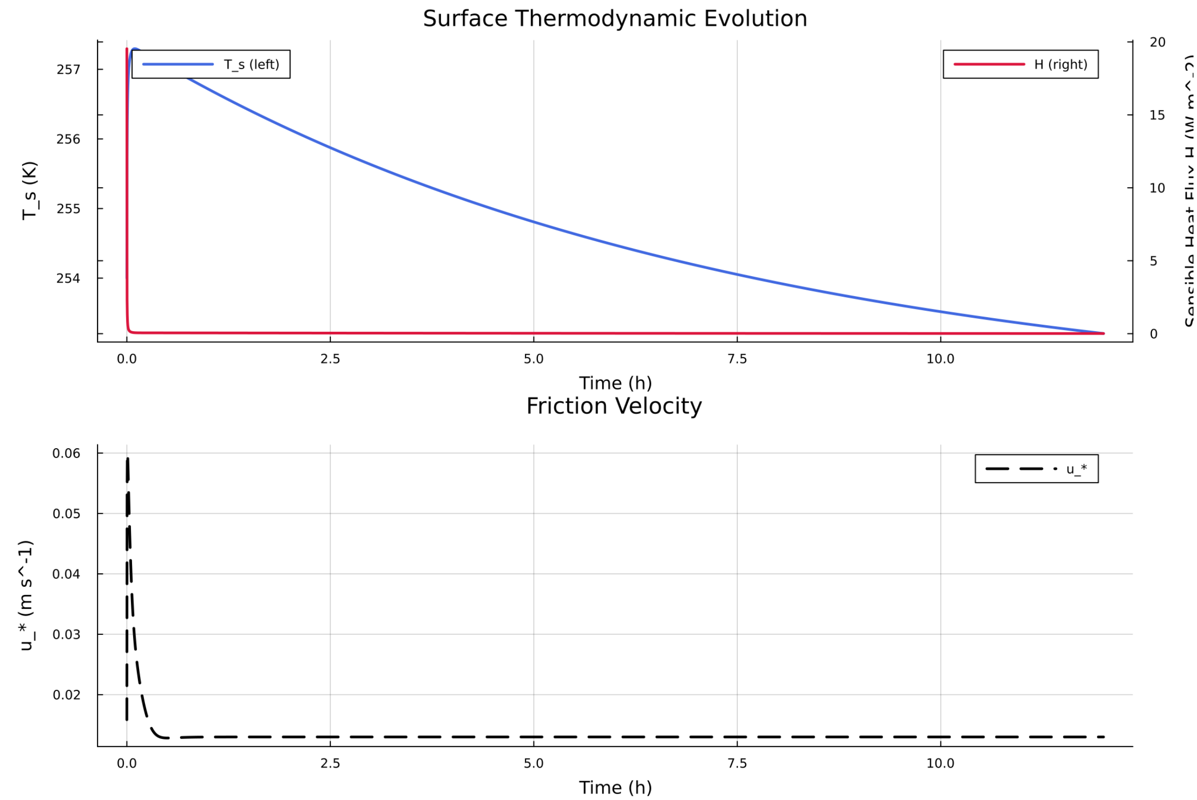

Figure 10: Time series of T__s, sensible heat flux, and friction velocity

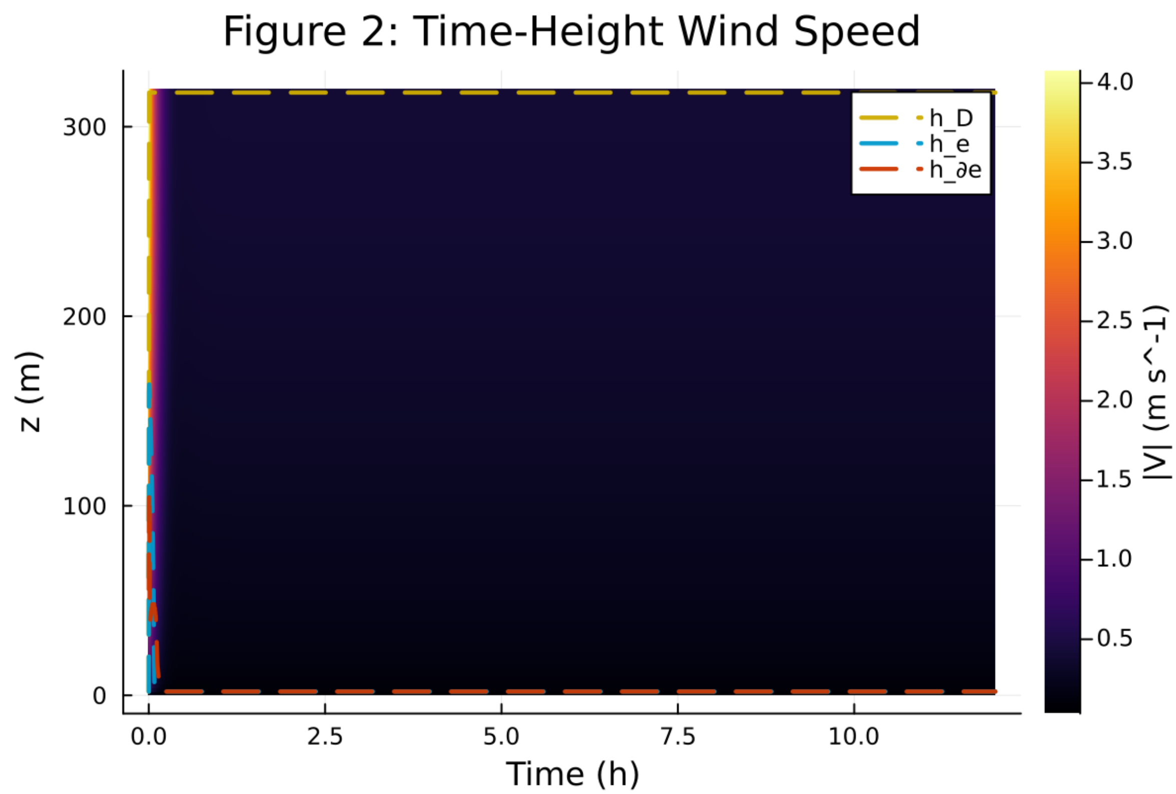

Figure 11: Time-height contour of wind speed with LLJ evolution

6 Case 3: gabls1 80x2m 9h report

7 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

7.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 101 / 0
- RHS evaluations: 25252
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

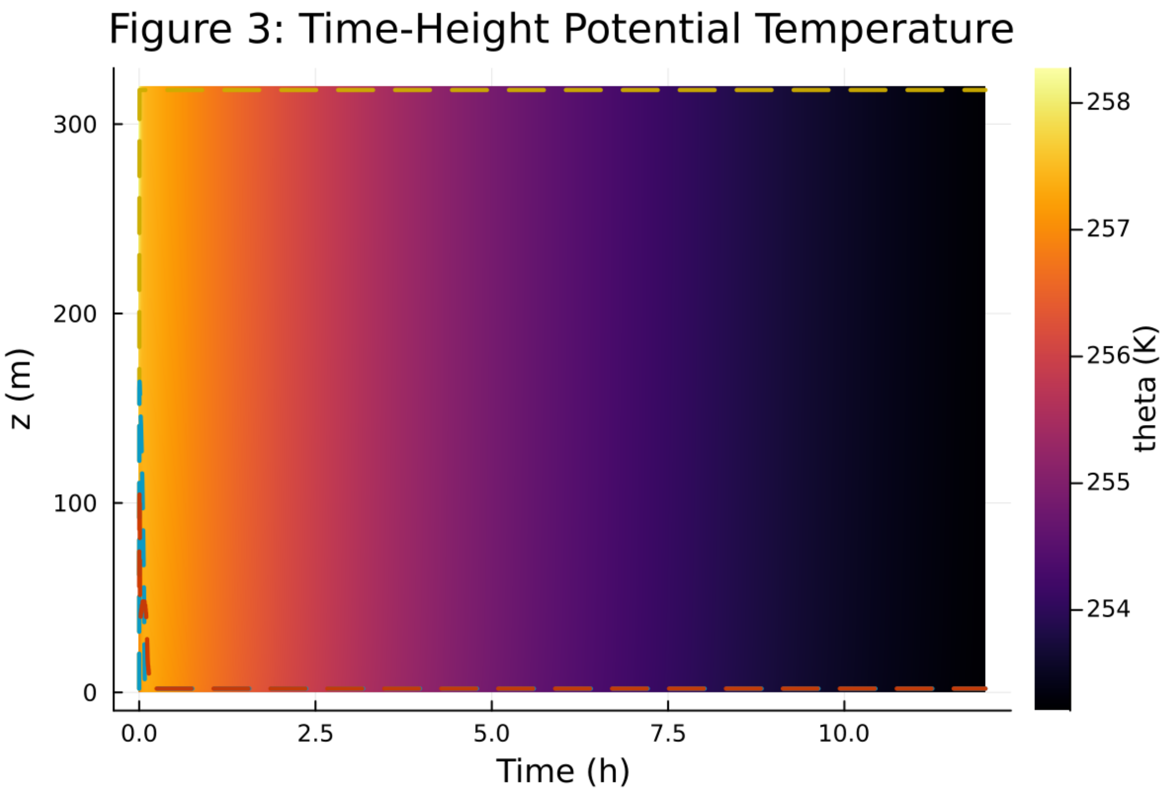

Figure 12: Time-height contour of potential temperature and inversion growth

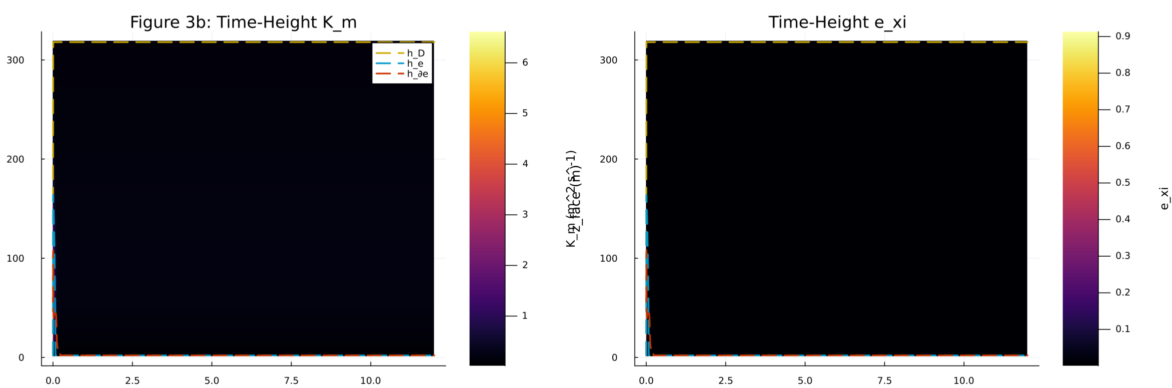

Figure 13: Vertical profiles of U, theta, and Km at representative times

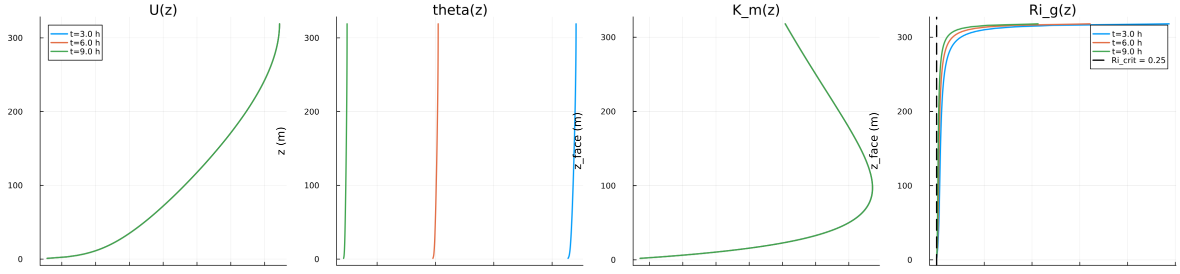

Figure 14: Surface energy budget components (R_n, H, G, storage)

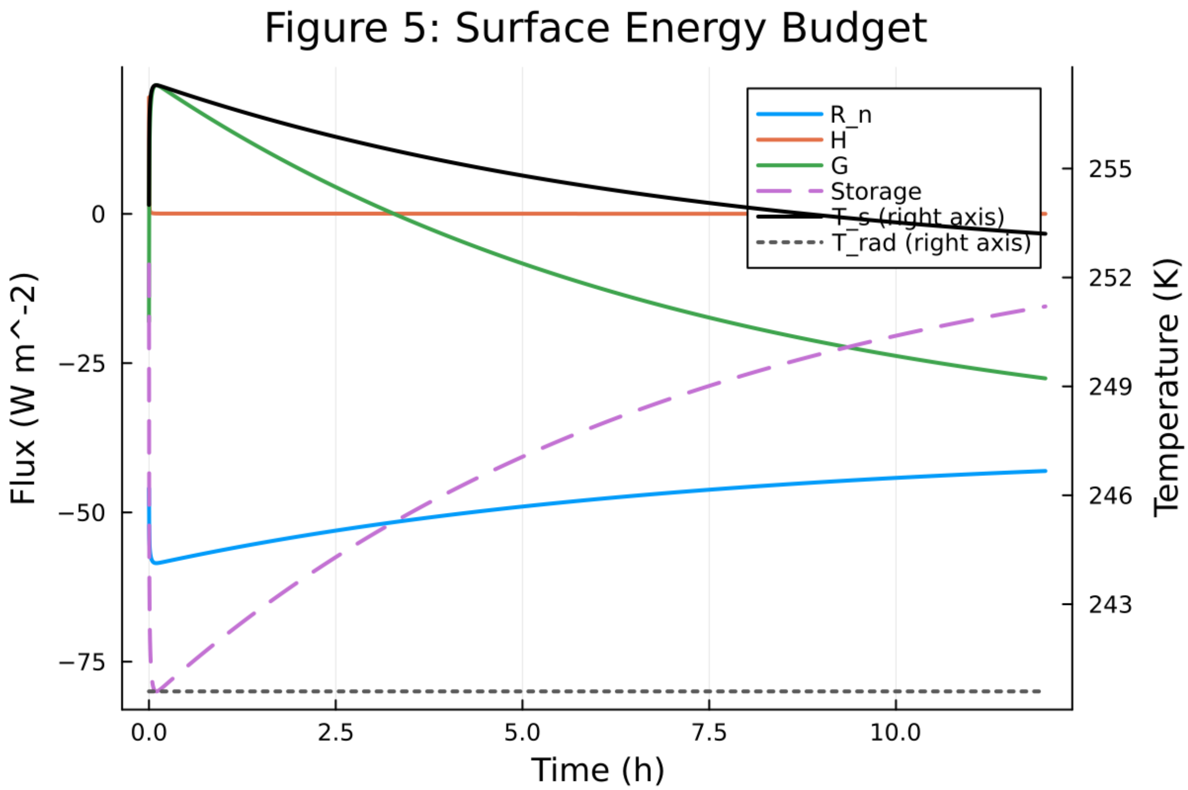

Figure 15: Phase portrait on regularized manifold (Delta vs e__xi)

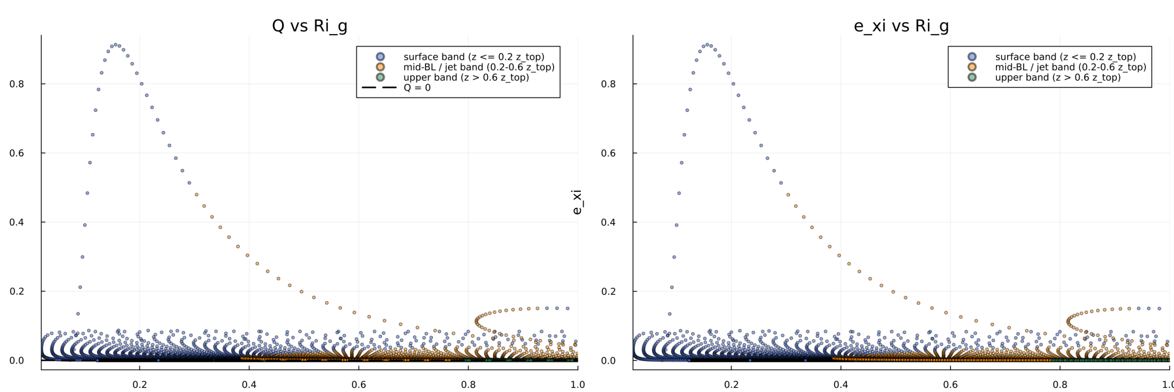

Figure 16: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

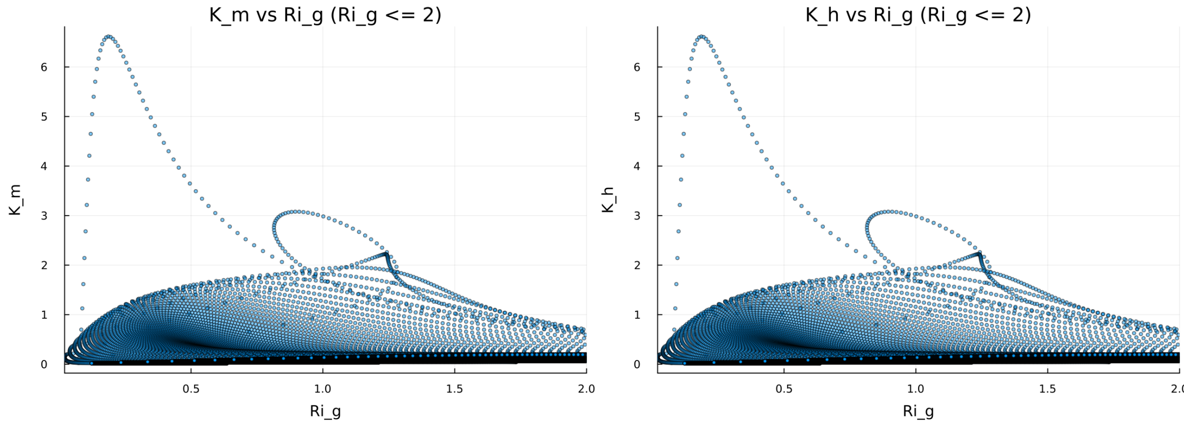

Figure 17: Fold proximity diagnostics versus time

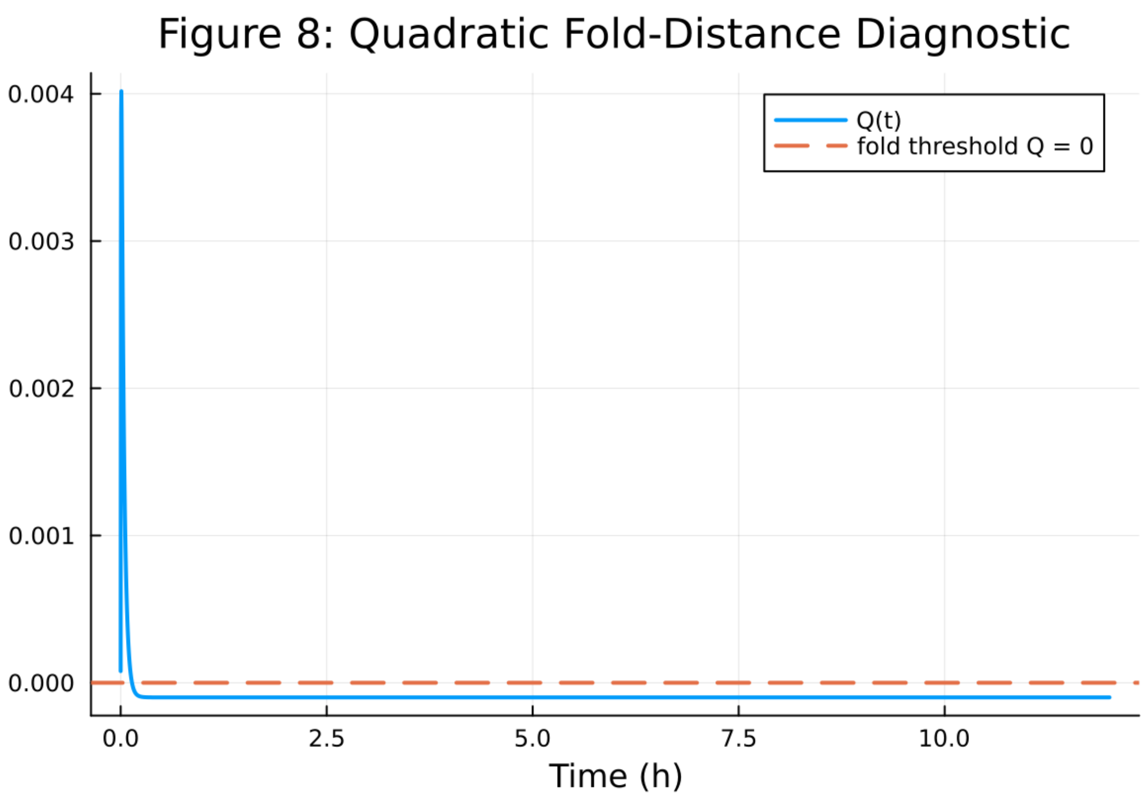

Figure 18: fig08 fold proximity.png

Table 3: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	8.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	0.100000 m
Surface roughness (heat)	z_{0h}	0.010000 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	245.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

7.2 Forcing and Boundary Conditions

7.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.092507%
- Diffusivity bounds: $K_m \in [0.017513, 3.564392]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.029775, 386.667515]$

7.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

7.5 Figure Manifest

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8. Fold proximity diagnostics versus time

7.6 Included Plots

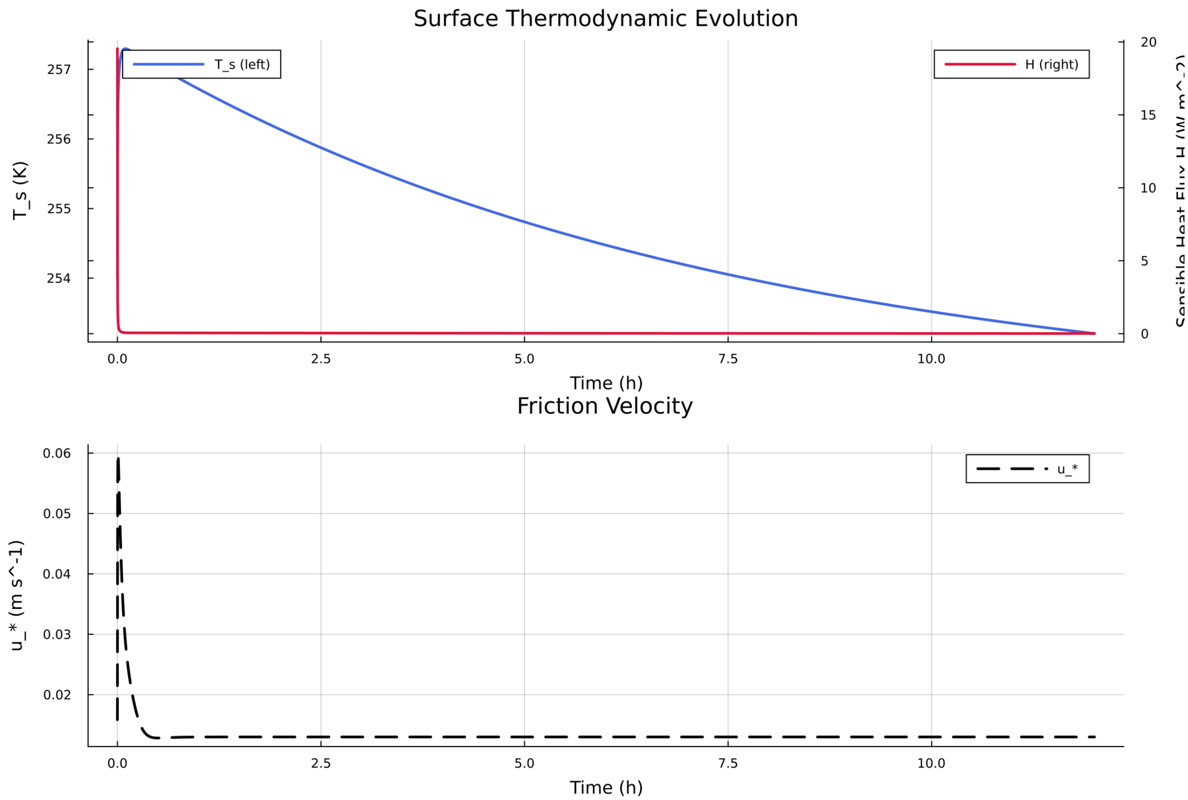

Figure 19: Time series of T__s, sensible heat flux, and friction velocity

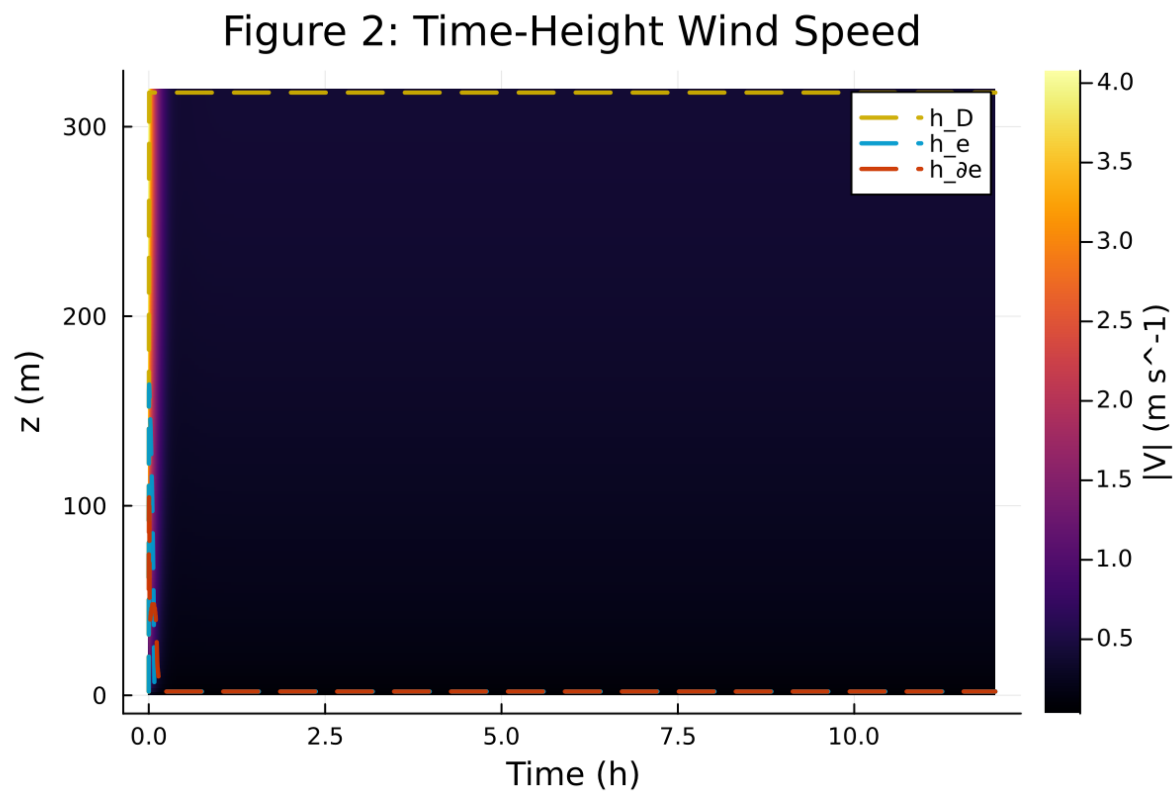

Figure 20: Time-height contour of wind speed with LLJ evolution

8 Case 4: gabls1 200x2m 12h report

9 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1_400m/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

9.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1_400m`
- Number of saved time samples: 1441
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 99 / 0
- RHS evaluations: 60392
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

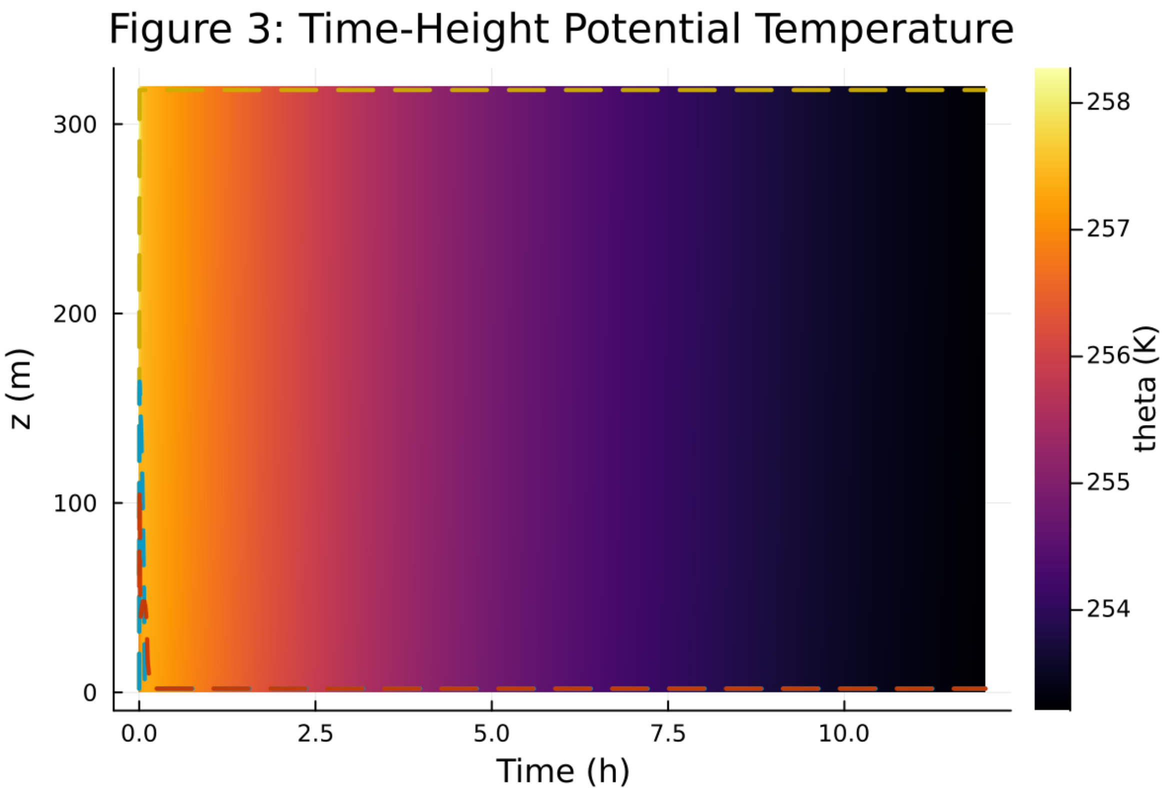

Figure 21: Time-height contour of potential temperature and inversion growth

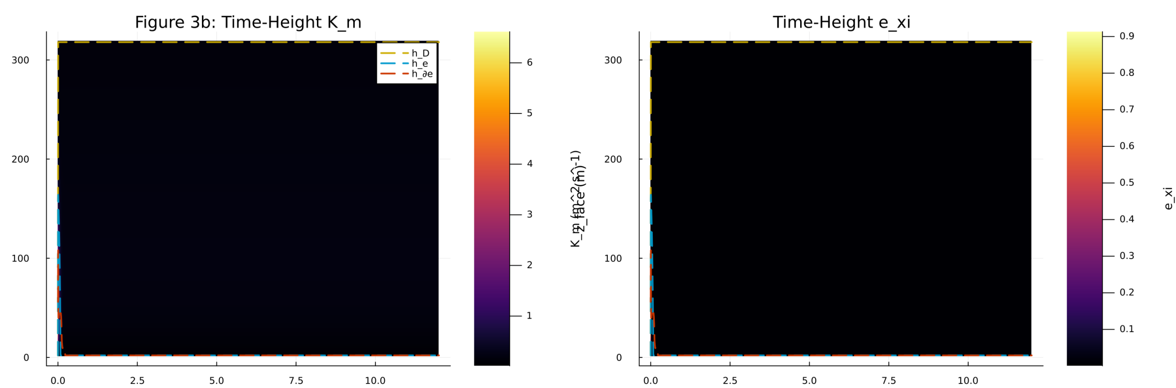

Figure 22: Vertical profiles of U, theta, and Km at representative times

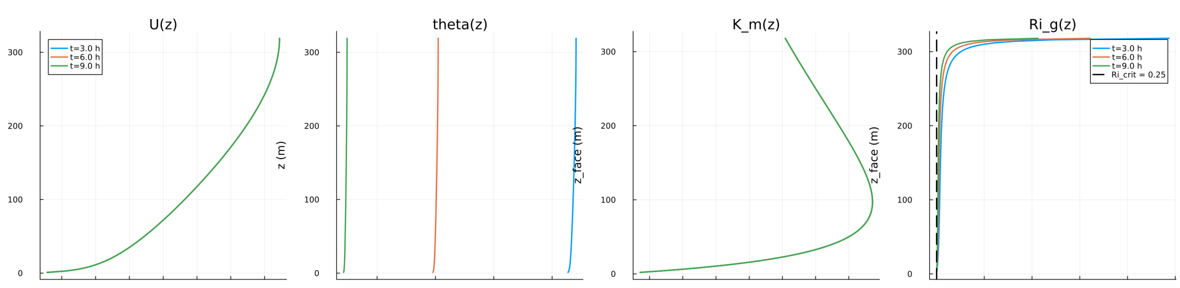

Figure 23: Surface energy budget components (R_n, H, G, storage)

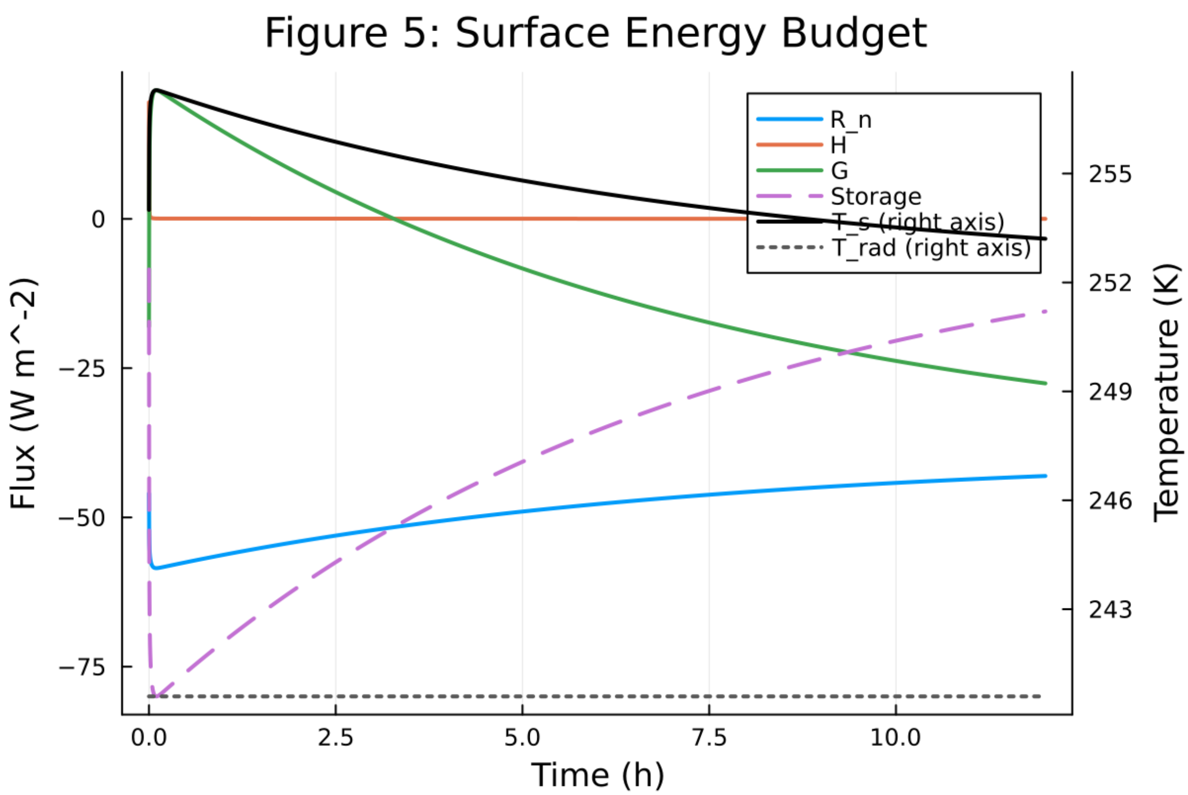

Figure 24: Phase portrait on regularized manifold (Delta vs e__xi)

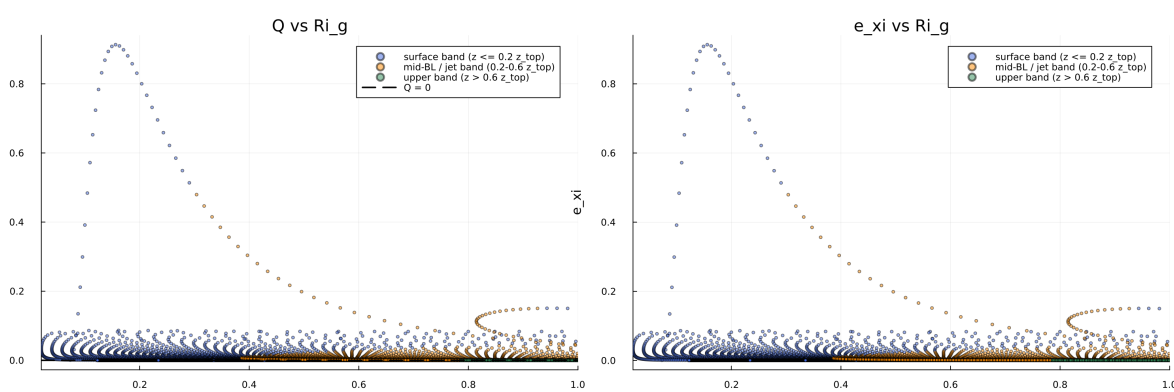

Figure 25: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

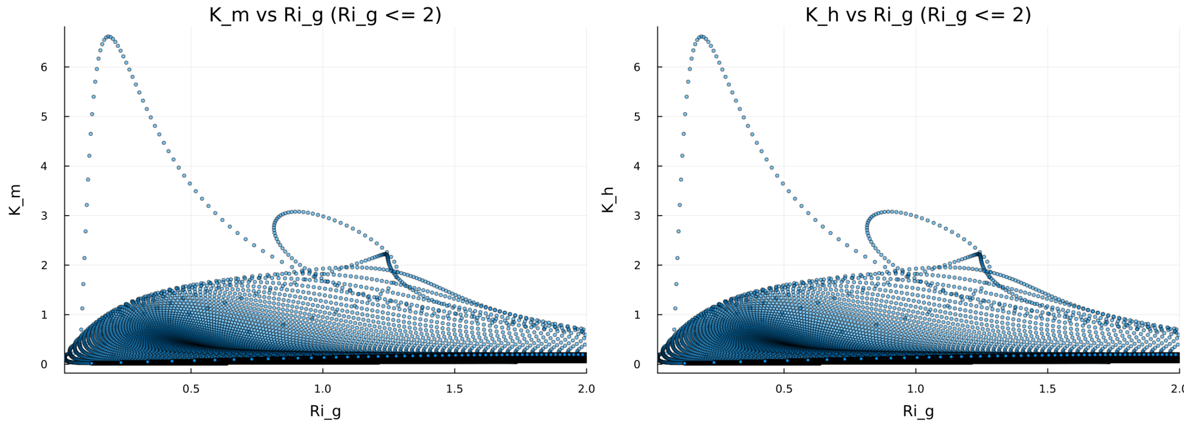

Figure 26: Fold proximity diagnostics versus time

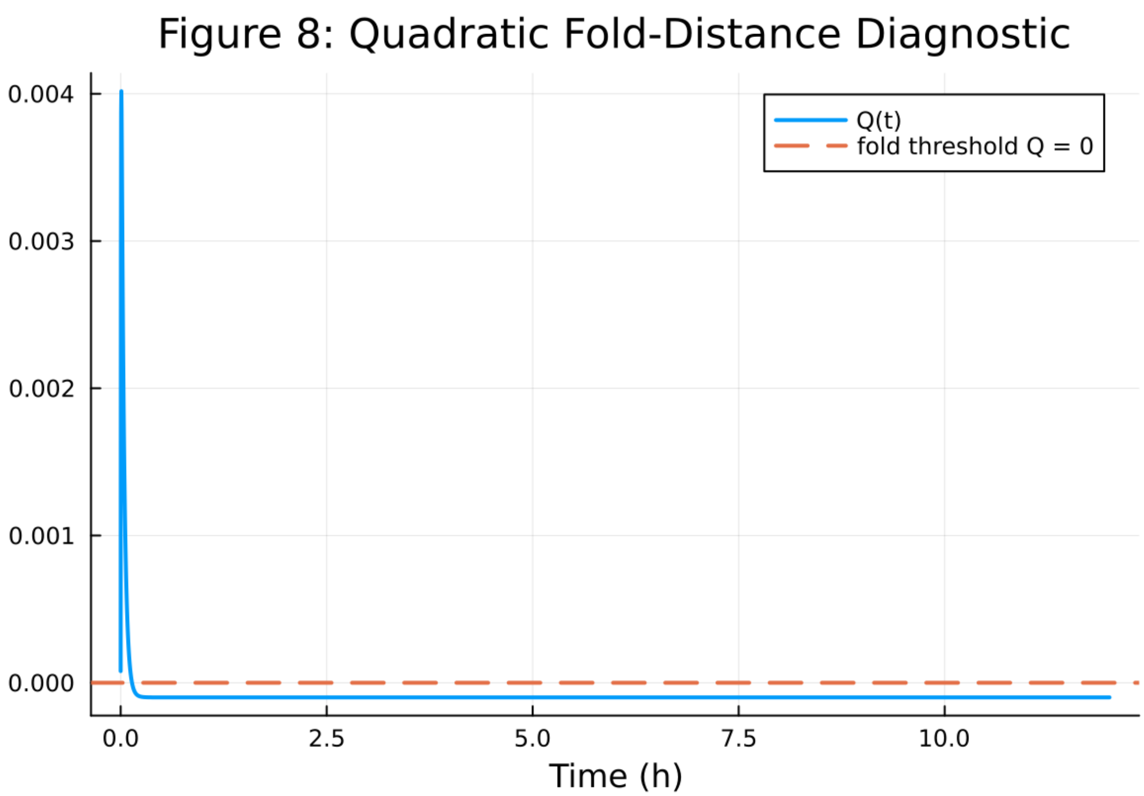

Figure 27: fig08 fold proximity.png

Table 4: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000\,\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	$0.000000\,\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	$0.100000\,\mathrm{m}$
Surface roughness (heat)	z_{0h}	$0.010000\,\mathrm{m}$
Coriolis parameter	f	$1.000\mathrm{e}{-04}\,\mathrm{s^{-1}}$
Reference potential temperature	θ_a	$265.000000\,\mathrm{K}$
Deep soil temperature	T_{deep}	$262.000000\,\mathrm{K}$
Downward longwave forcing	$R_\downarrow$	$245.000000\,\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	$0.100000\,\mathrm{m}$
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	$0.001000\,\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	$265.000000\,\mathrm{K}$
Top relaxation coefficient	λ_{top}	$0.000000\,\mathrm{s^{-1}}$

9.2 Forcing and Boundary Conditions

9.3 Numerical Verification

- Maximum surface energy closure error: $0\,\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.346981%
- Diffusivity bounds: $K_m \in [0.006021, 3.564392]\ \mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.031448, 843.105664]$

9.4 Generated Artifacts

- Time-series CSV: `results/gabls1_400m/time_series.csv`
- Diagnostic summary JSON: `results/gabls1_400m/summary.json`
- Payload JLD2: `results/gabls1_400m/payload.jld2`

9.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
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6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

9.6 Included Plots

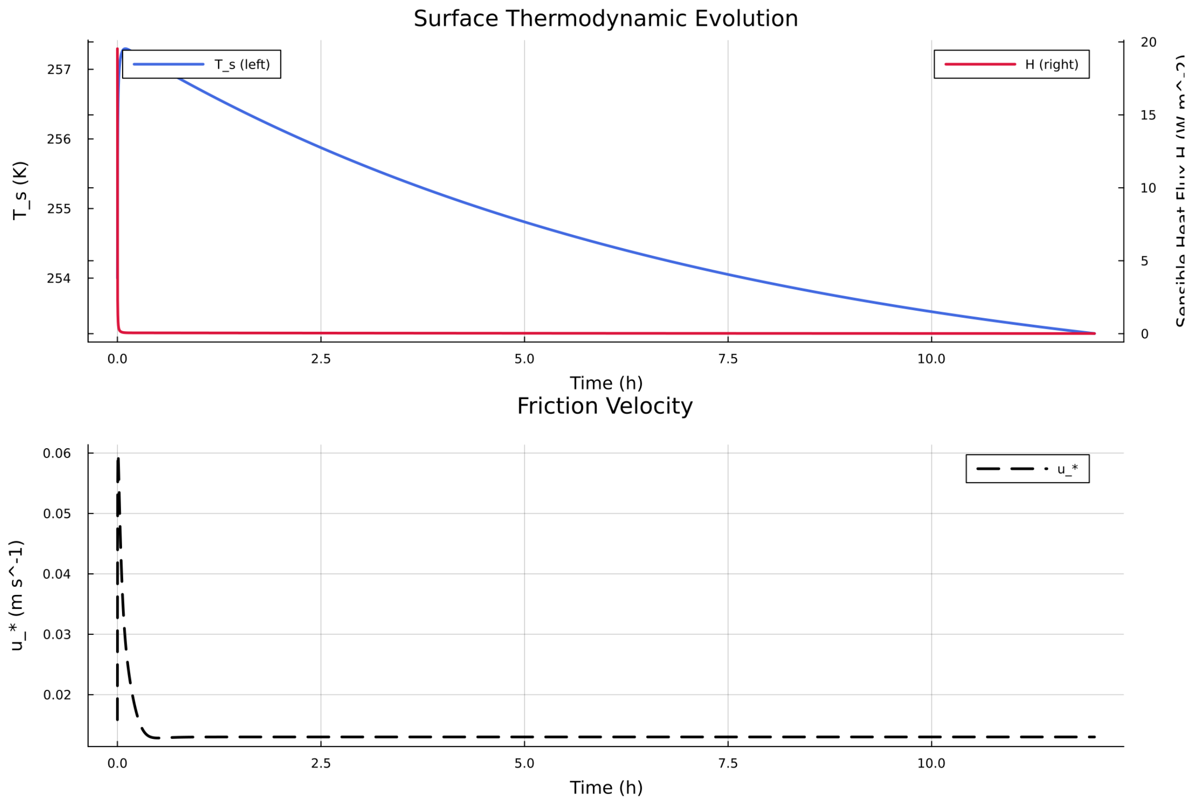

Figure 28: Time series of T__s, sensible heat flux, and friction velocity

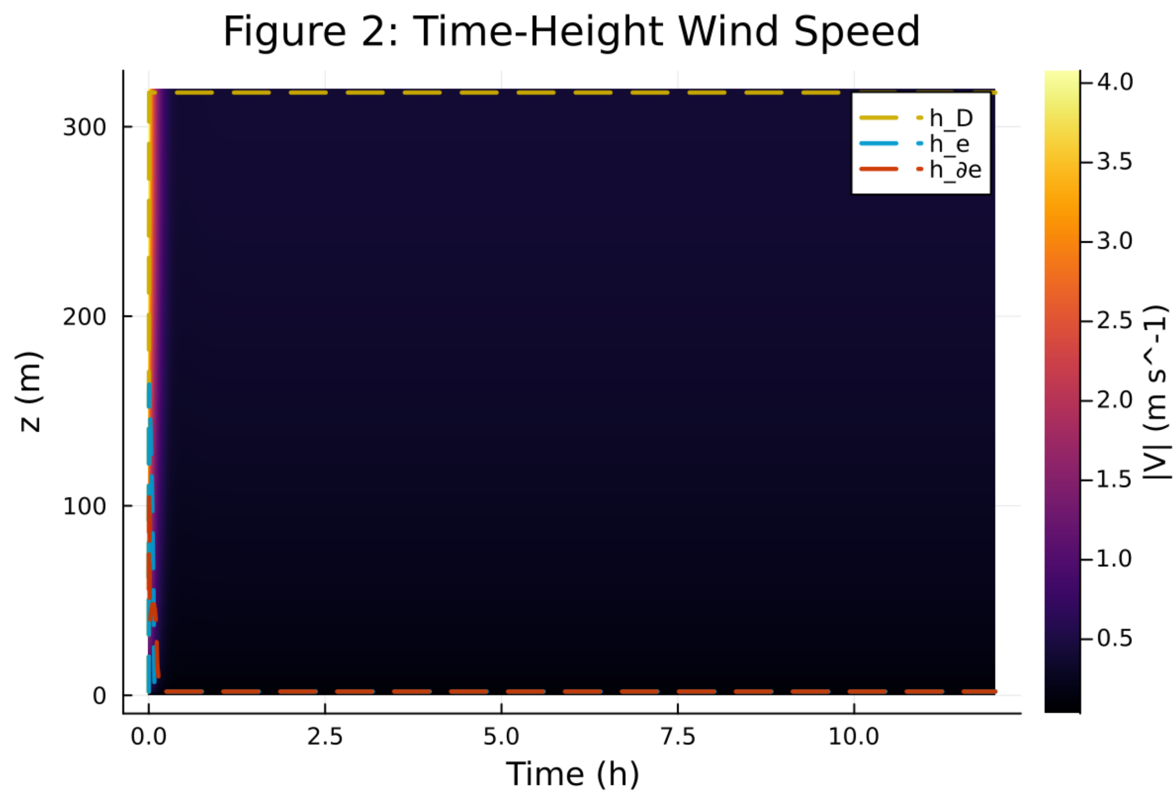

Figure 29: Time-height contour of wind speed with LLJ evolution

10 Case 5: idealized sbl 80x2m 9h report

11 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

11.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 106 / 0
- RHS evaluations: 26502
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

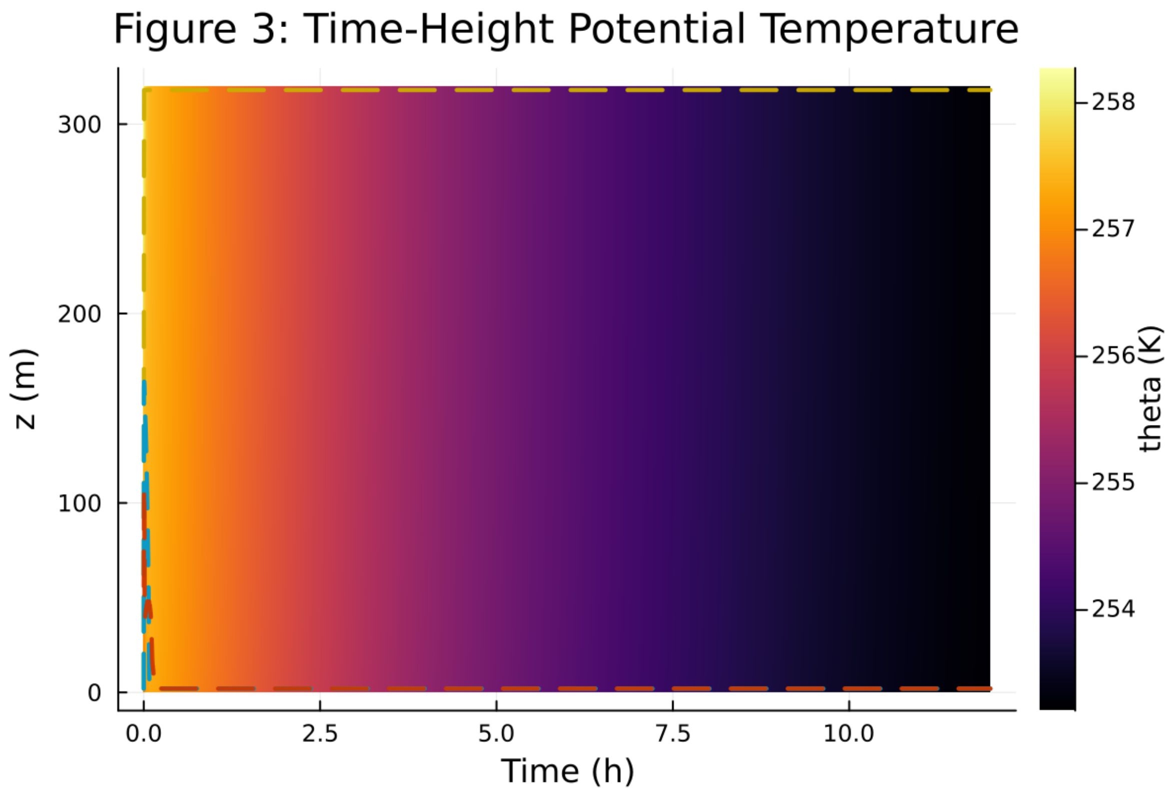

Figure 30: Time-height contour of potential temperature and inversion growth

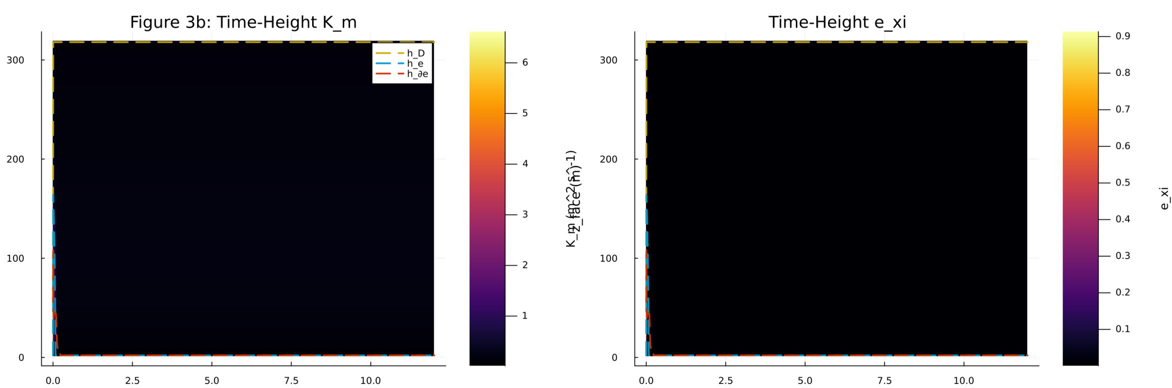

Figure 31: Vertical profiles of U, theta, and Km at representative times

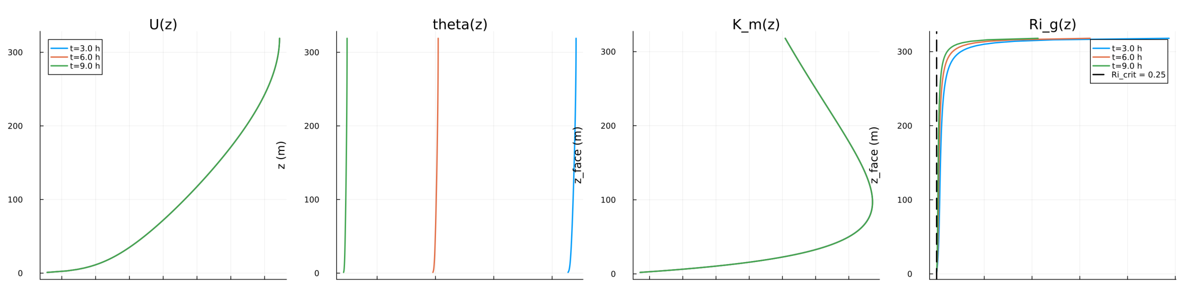

Figure 32: Surface energy budget components (R_n, H, G, storage)

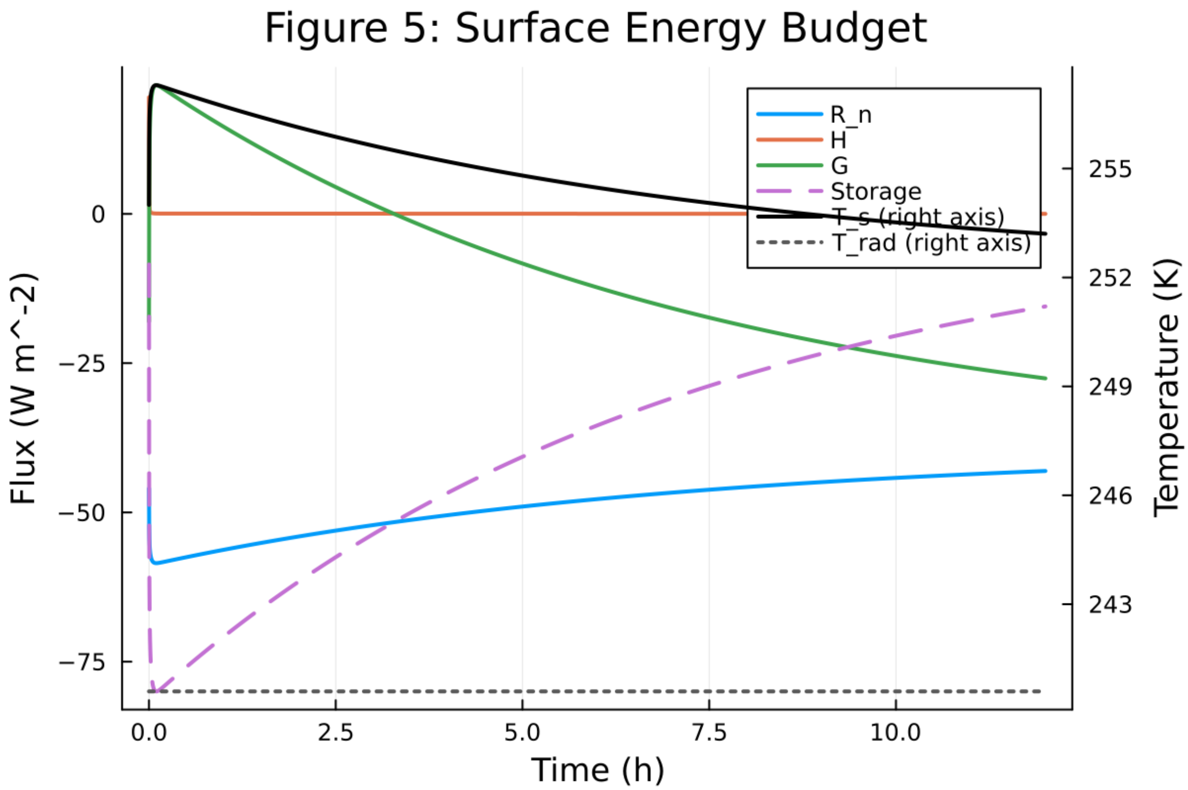

Figure 33: Phase portrait on regularized manifold (Delta vs e__xi)

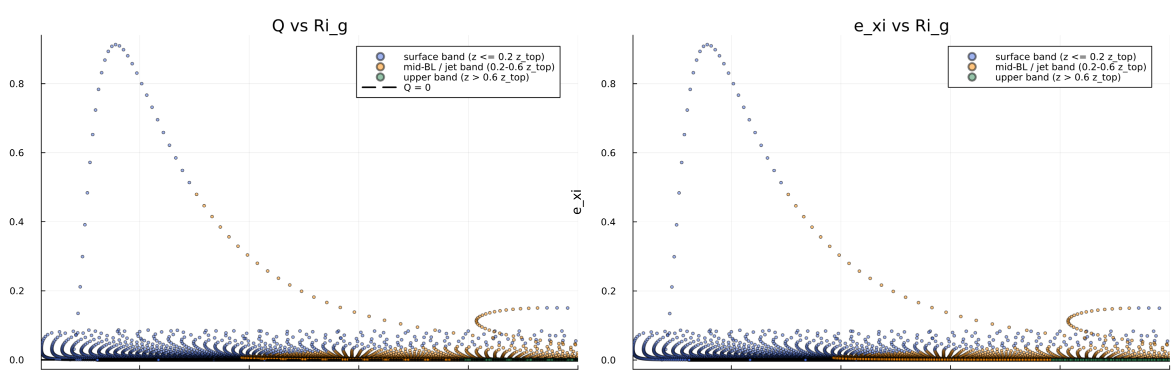

Figure 34: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

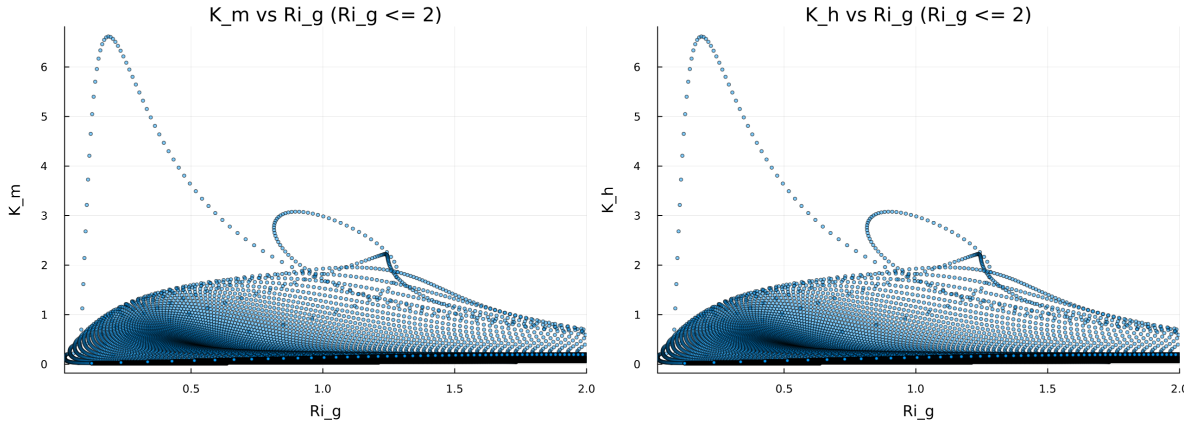

Figure 35: Fold proximity diagnostics versus time

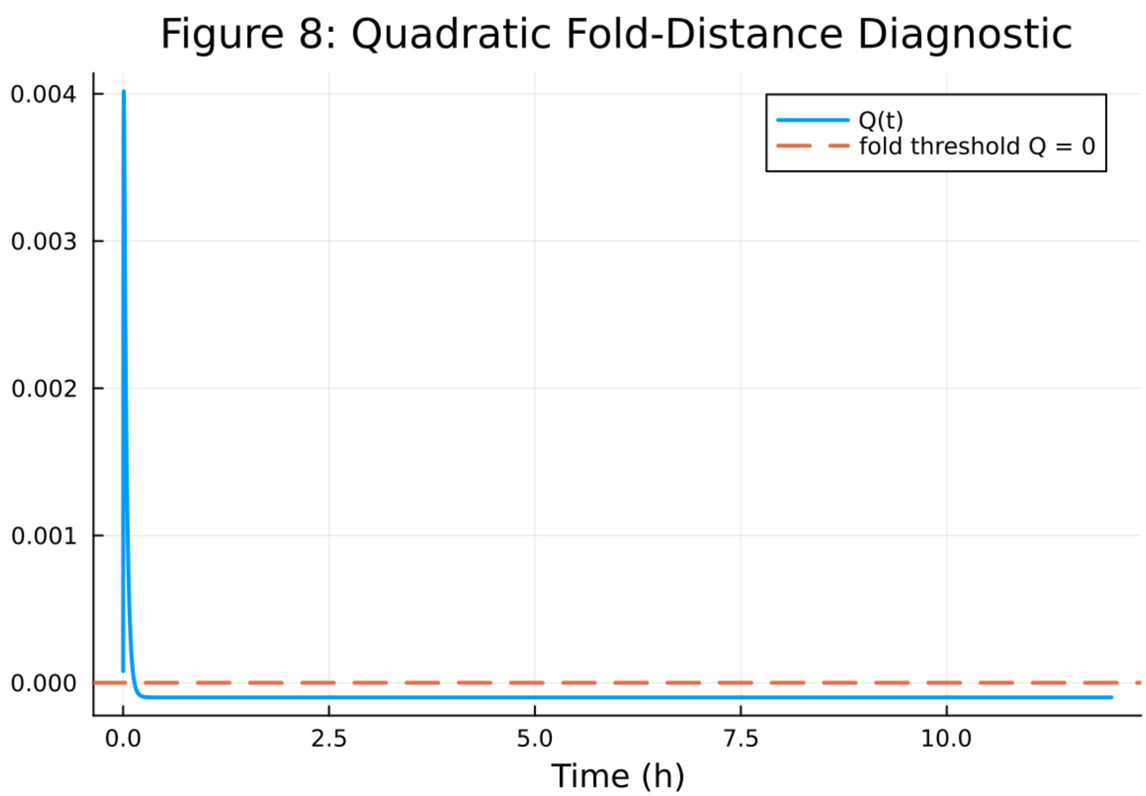

Figure 36: fig08 fold proximity.png

Table 5: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	10.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_\downarrow$	300.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

11.2 Forcing and Boundary Conditions

11.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.092507%
- Diffusivity bounds: $K_m \in [0.017515, 9.529353]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.011071, 348.796639]$

11.4 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

11.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

11.6 Included Plots

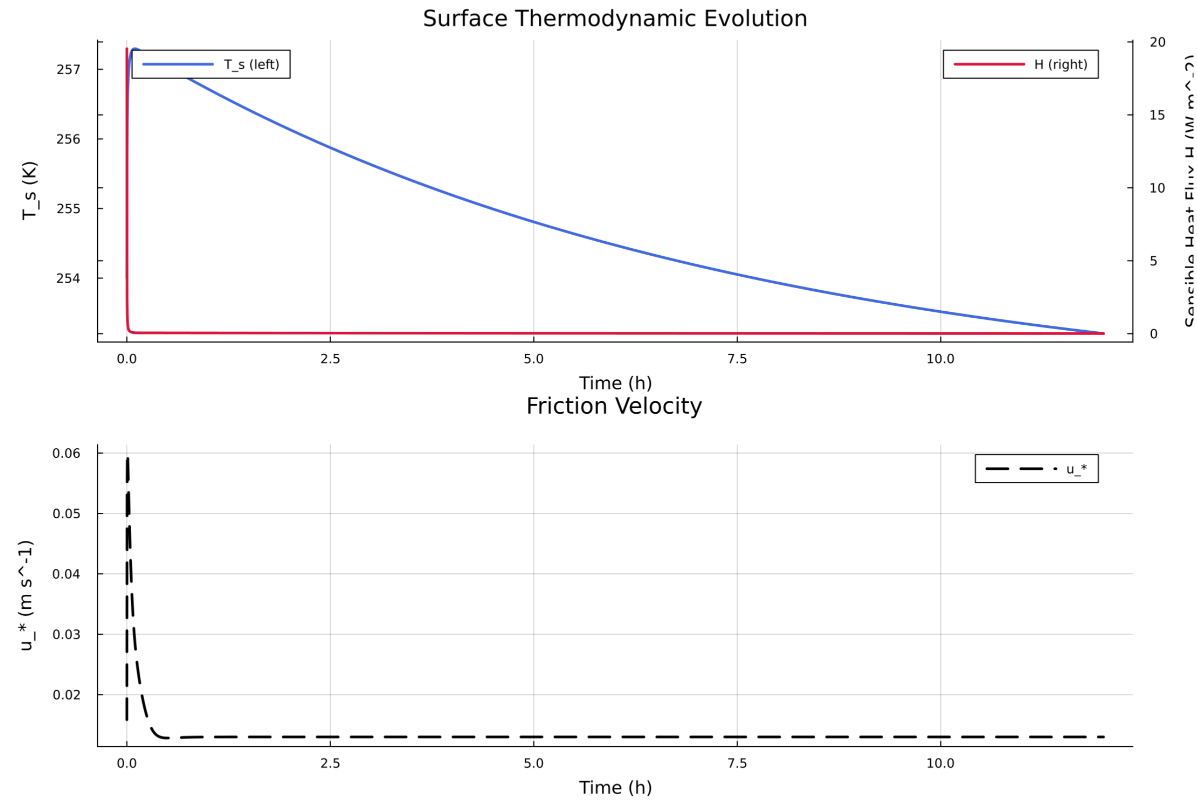

Figure 37: Time series of T__s, sensible heat flux, and friction velocity

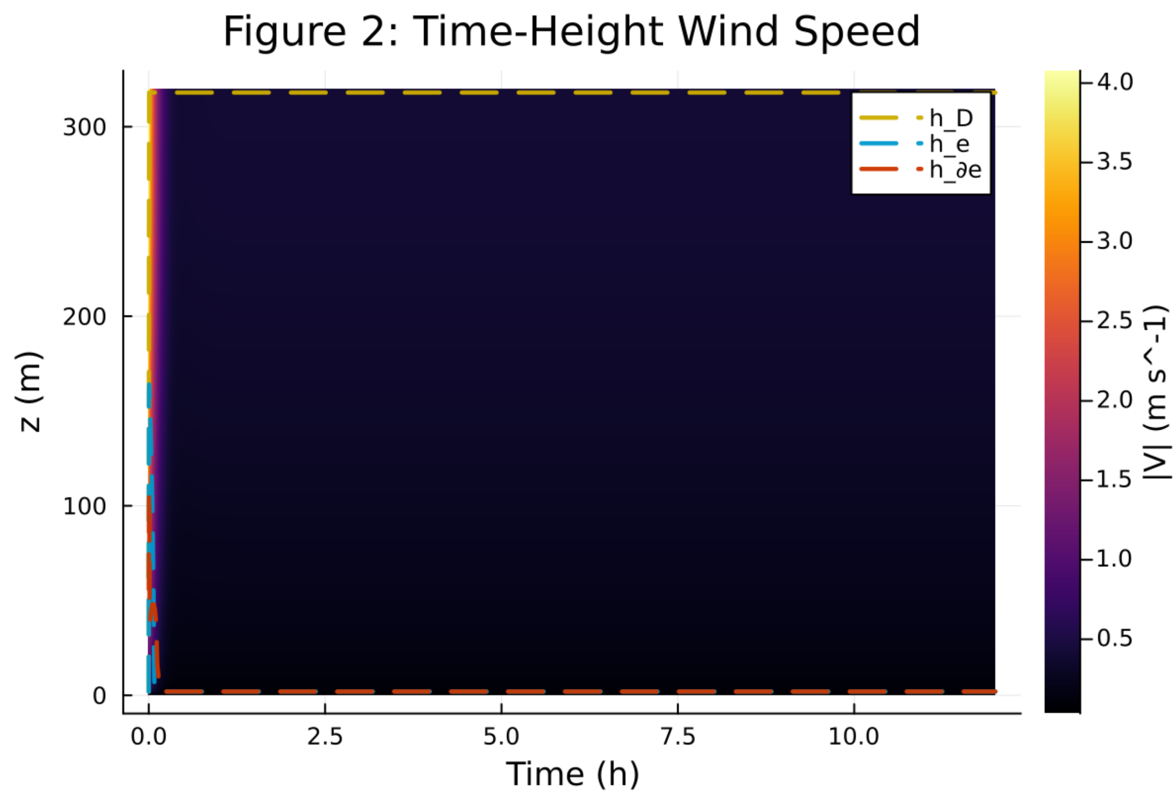

Figure 38: Time-height contour of wind speed with LLJ evolution

12 Case 6: scm verify 40x4m 0h report

13 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

13.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 52 / 0
- RHS evaluations: 6762
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

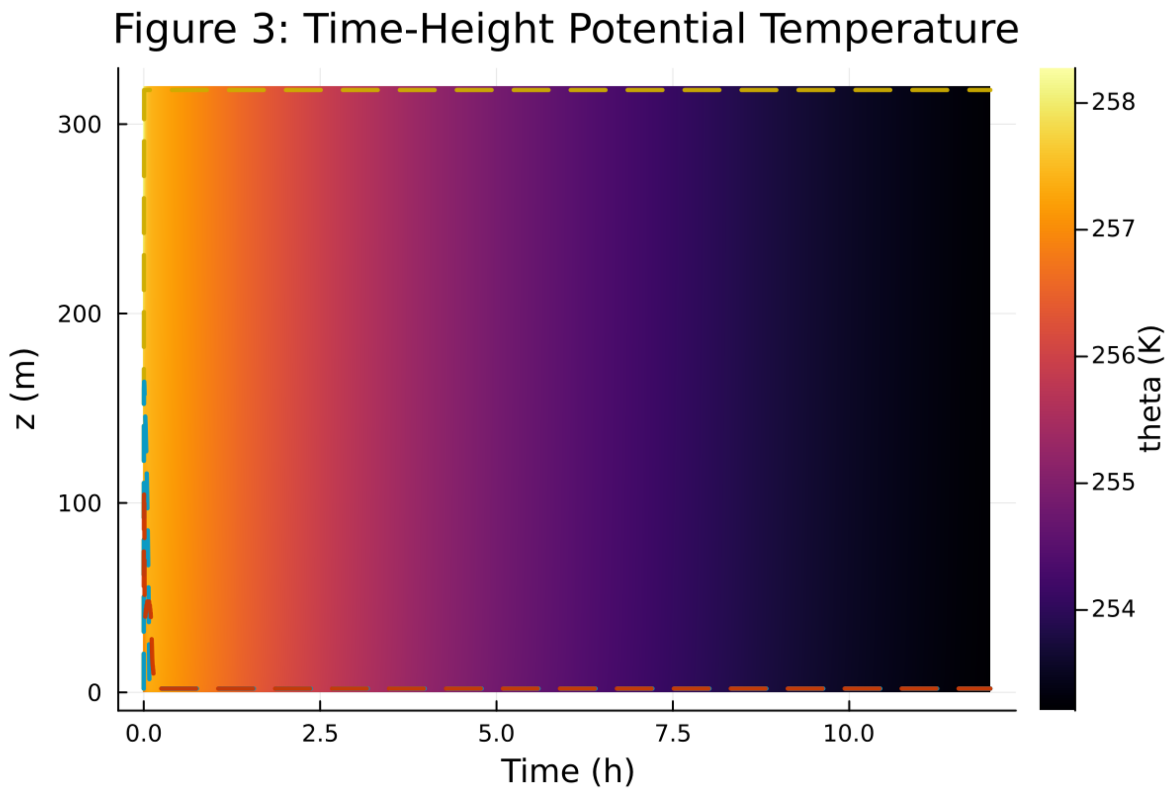

Figure 39: Time-height contour of potential temperature and inversion growth

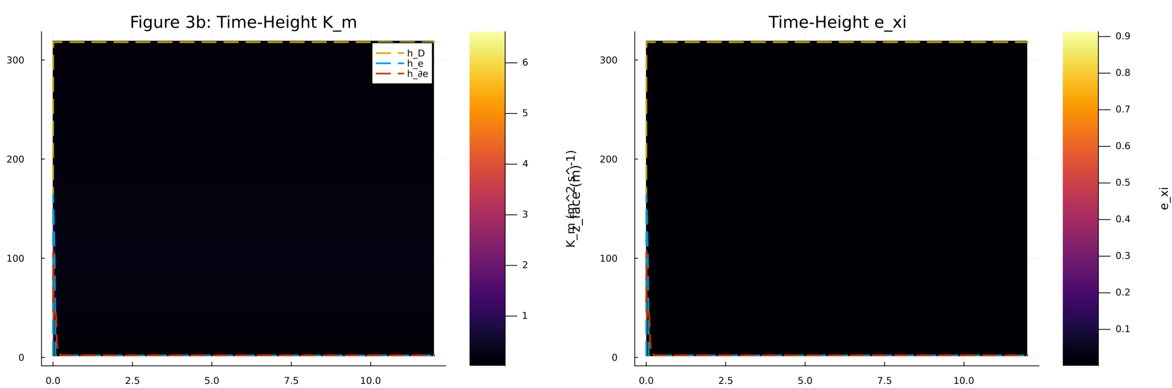

Figure 40: Vertical profiles of U, theta, and Km at representative times

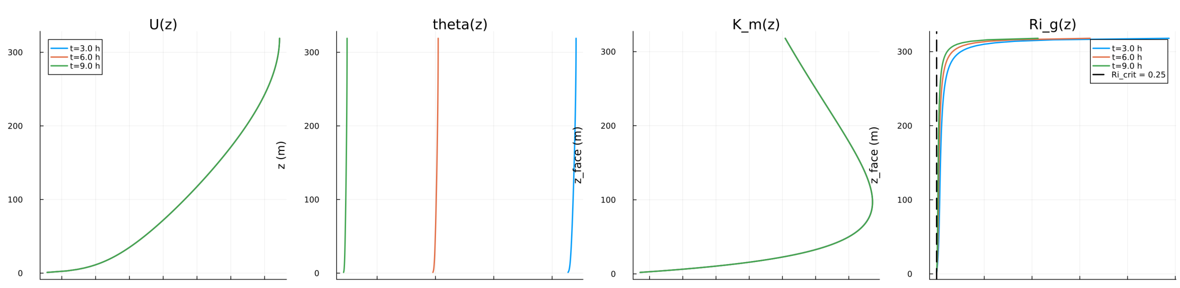

Figure 41: Surface energy budget components (R_n, H, G, storage)

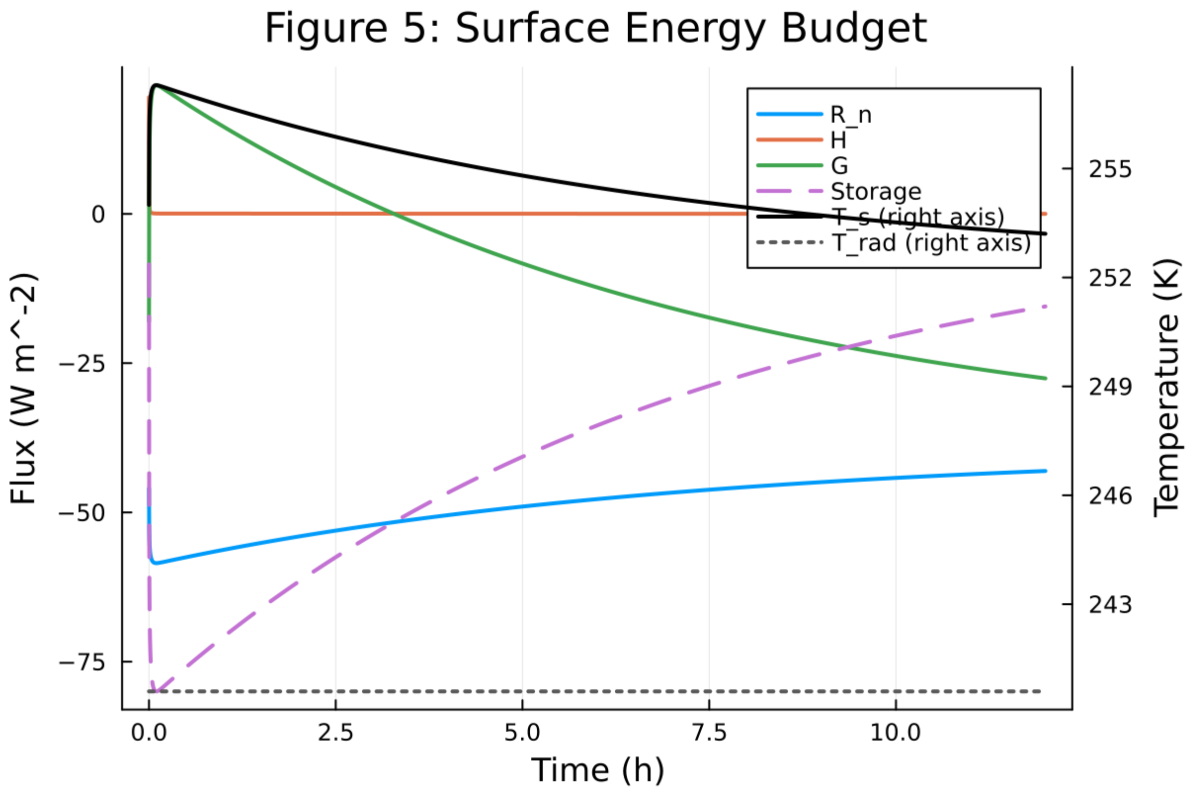

Figure 42: Phase portrait on regularized manifold (Delta vs e__xi)

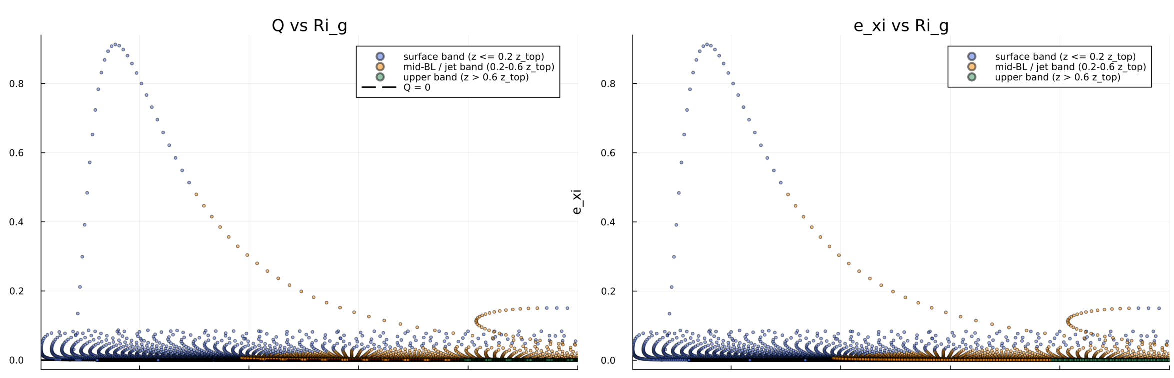

Figure 43: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

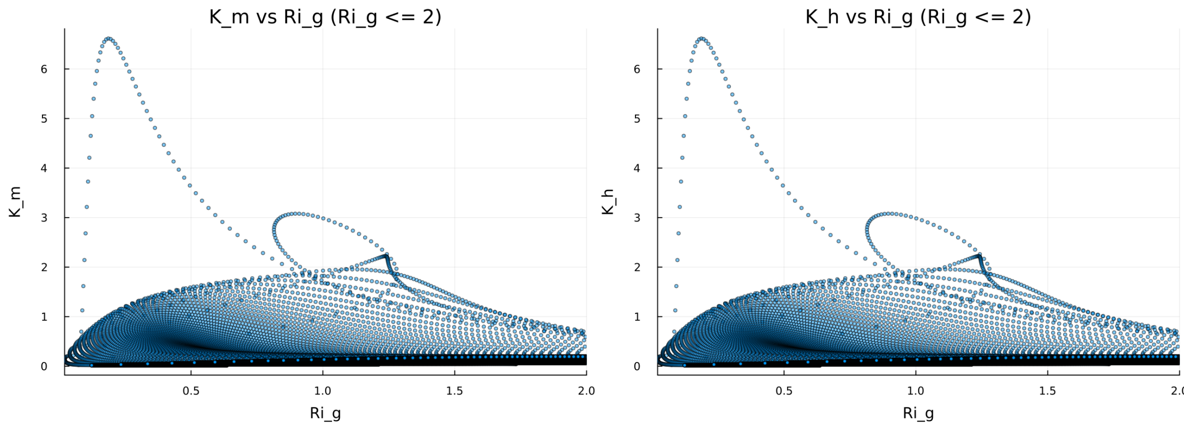

Figure 44: Fold proximity diagnostics versus time

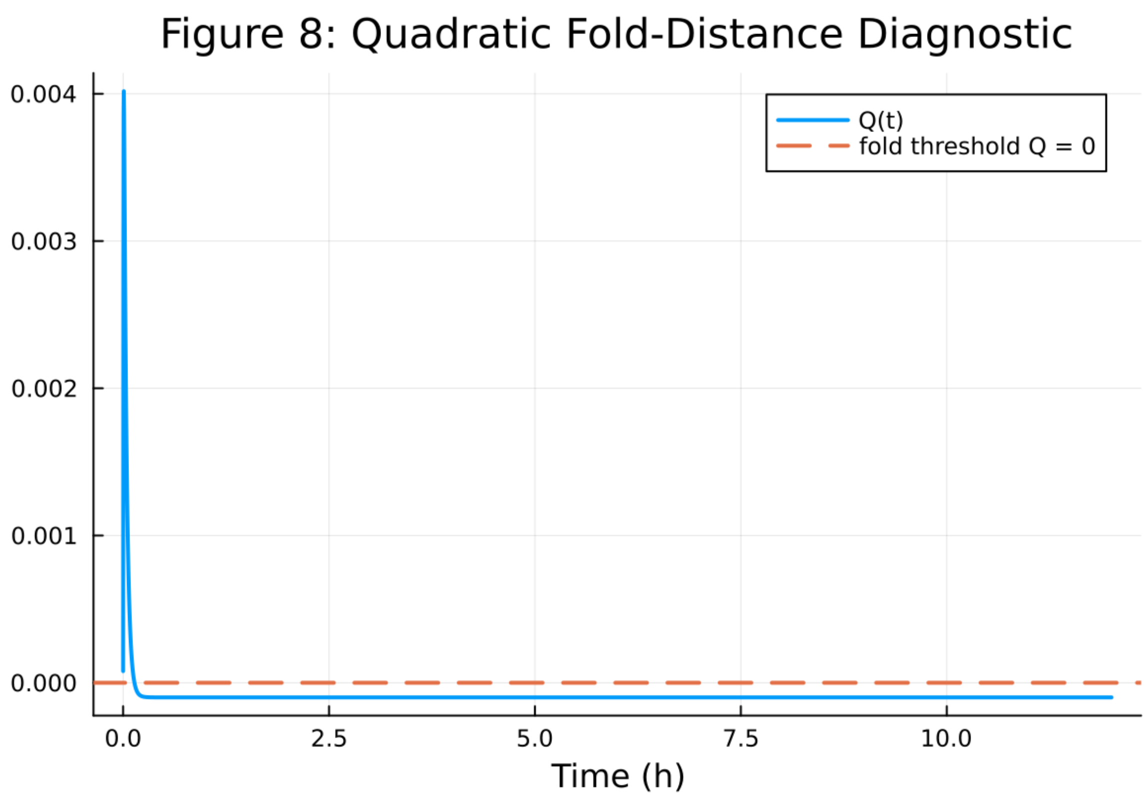

Figure 45: fig08 fold proximity.png

Table 6: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	10.000000 m s^{-1}
Meridional geostrophic wind	V_g	0.000000 m s^{-1}
Surface roughness (momentum)	z_{0m}	0.020000 m
Surface roughness (heat)	z_{0h}	0.005000 m
Coriolis parameter	f	1.000e-04 s^{-1}
Reference potential temperature	θ_a	280.000000 K
Deep soil temperature	T_{deep}	276.000000 K
Downward longwave forcing	$R_\downarrow$	300.000000 W m^{-2}
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 m^2 s^{-1}
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

13.2 Forcing and Boundary Conditions

13.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 7.692308%
- Diffusivity bounds: $K_m \in [0.035803, 9.530310]$ m^2 s^{-1}
- Richardson number bounds: $Ri \in [0.019320, 10.640107]$

13.4 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

13.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

13.6 Included Plots

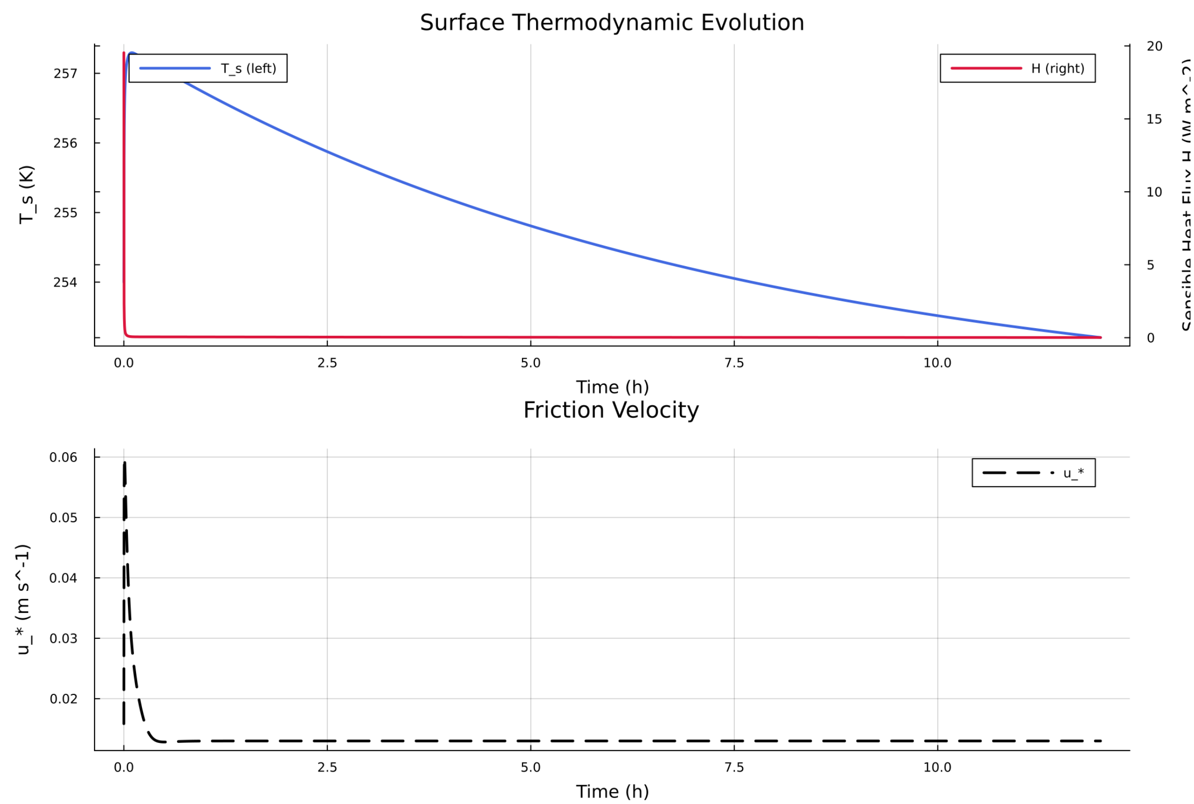

Figure 46: Time series of T__s, sensible heat flux, and friction velocity

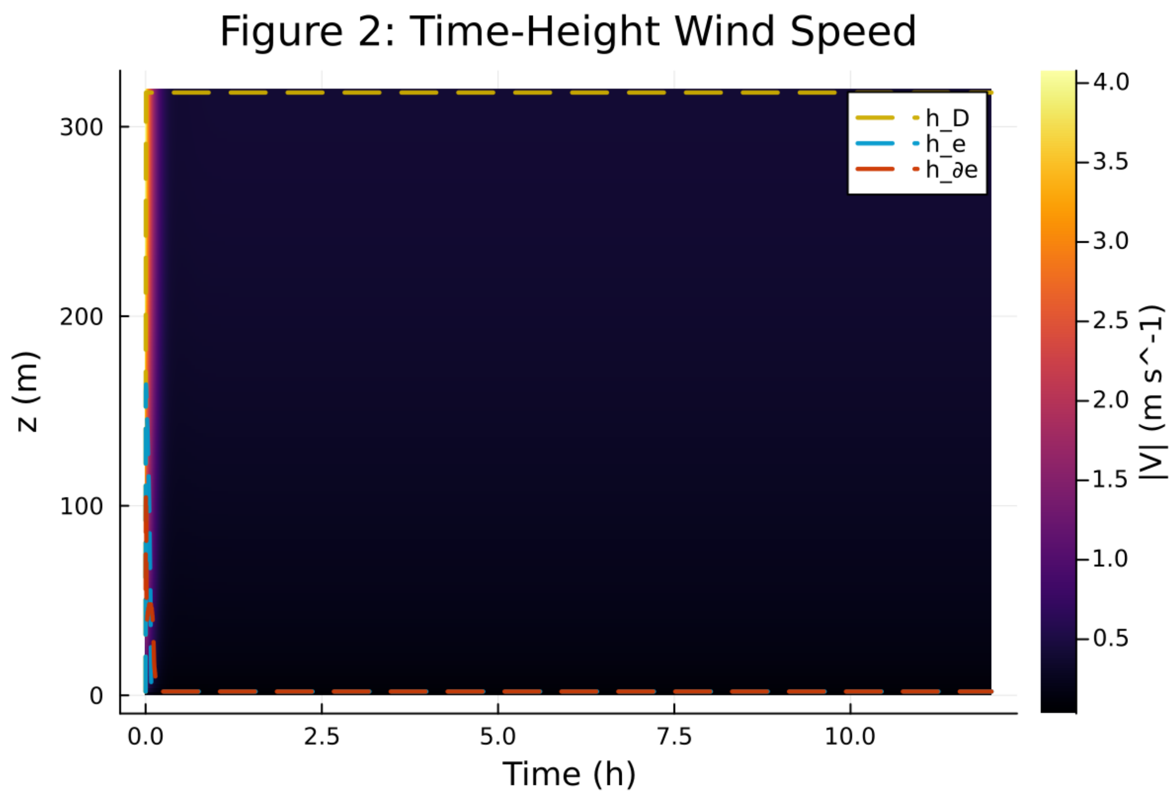

Figure 47: Time-height contour of wind speed with LLJ evolution

14 Case 7: sheba 250x2m 12h report

15 SCM Case Summary: sheba

This section was generated automatically from the SCM diagnostics artifact `results/sheba_high_top/summary.json` . The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoForwardDiff{nothing, Nothing}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRECS), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success` .

15.1 Core Run Metadata

- Case identifier: `sheba`
- Output directory: `results/sheba_high_top`
- Number of saved time samples: 4321
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 97 / 0
- RHS evaluations: 1748
- Solver tolerances: abstol $= 1.000 \times 10^{-8}$, reltol $= 1.000 \times 10^{-6}$

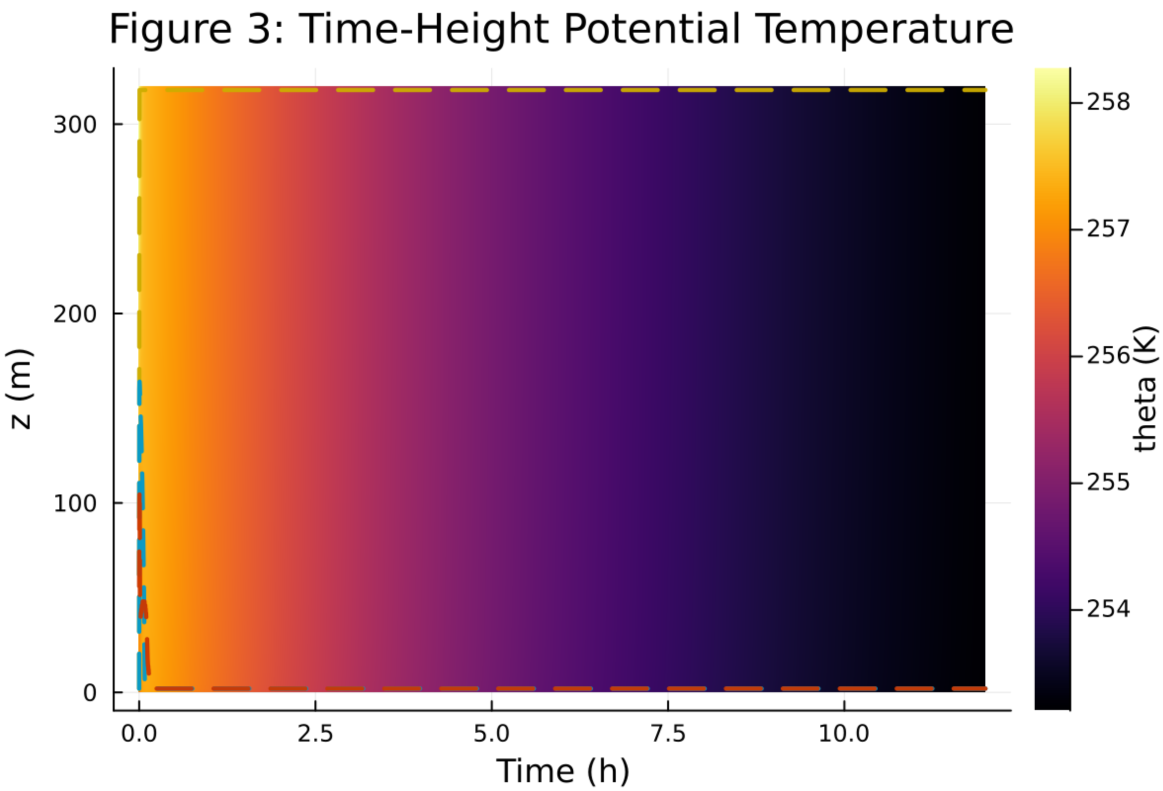

Figure 48: Time-height contour of potential temperature and inversion growth

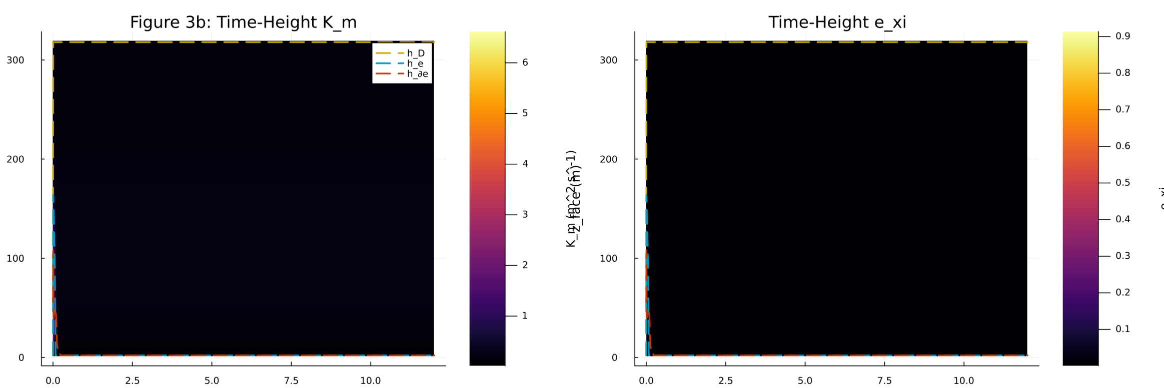

Figure 49: Vertical profiles of U, theta, and Km at representative times

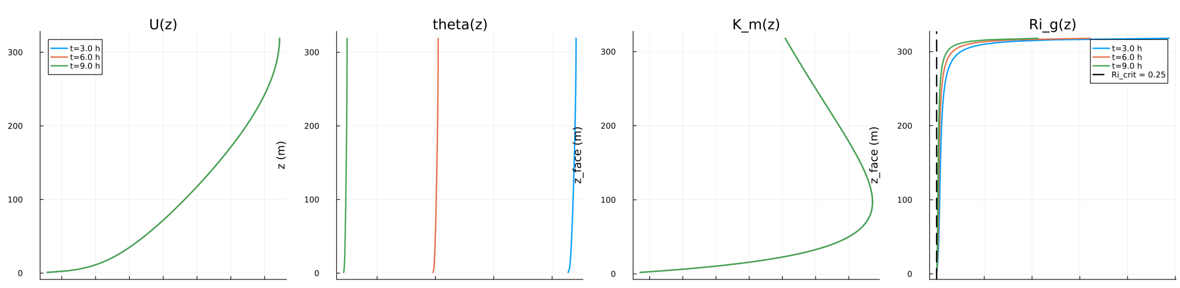

Figure 50: Surface energy budget components (R_n, H, G, storage)

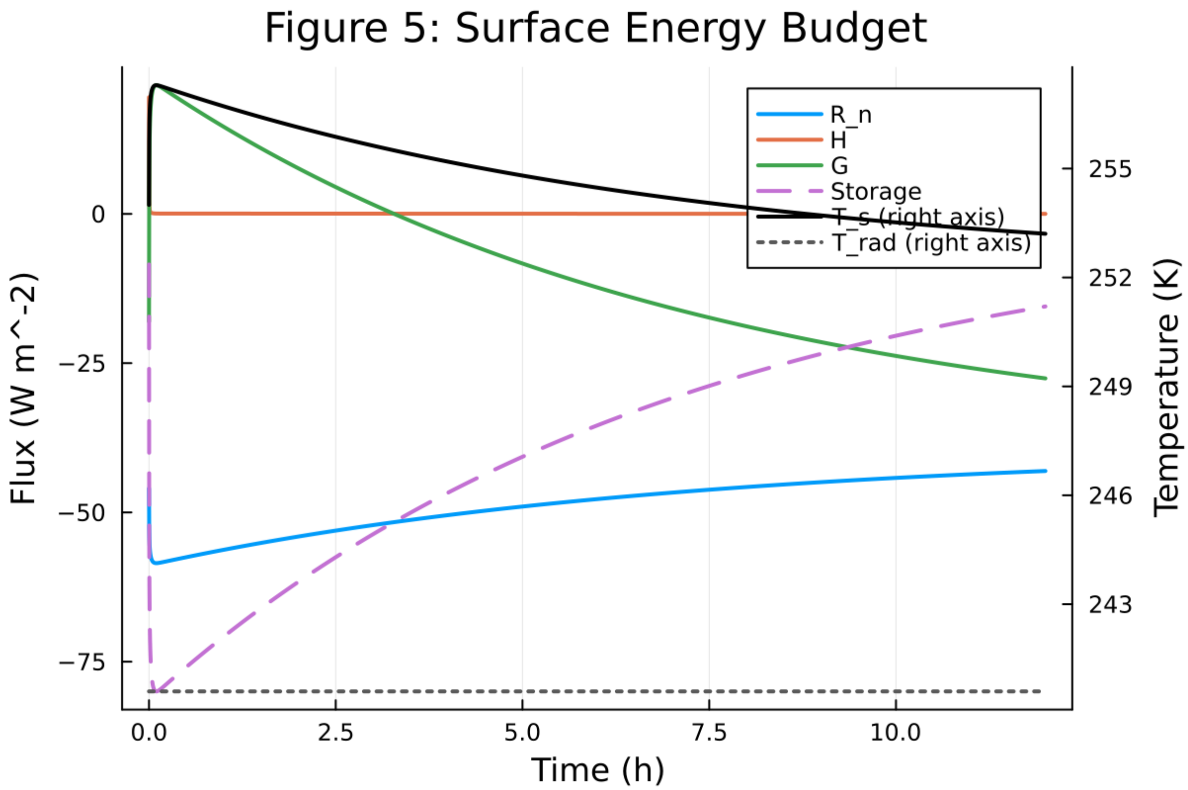

Figure 51: Phase portrait on regularized manifold (Delta vs e__xi)

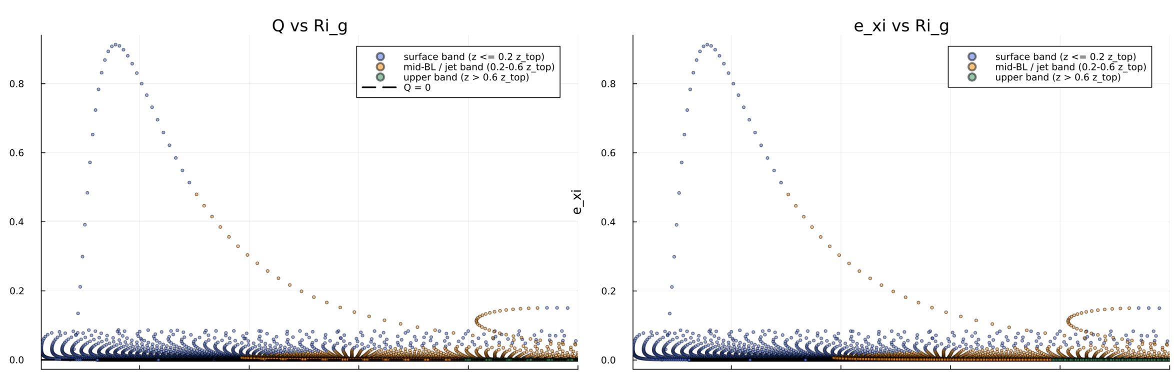

Figure 52: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

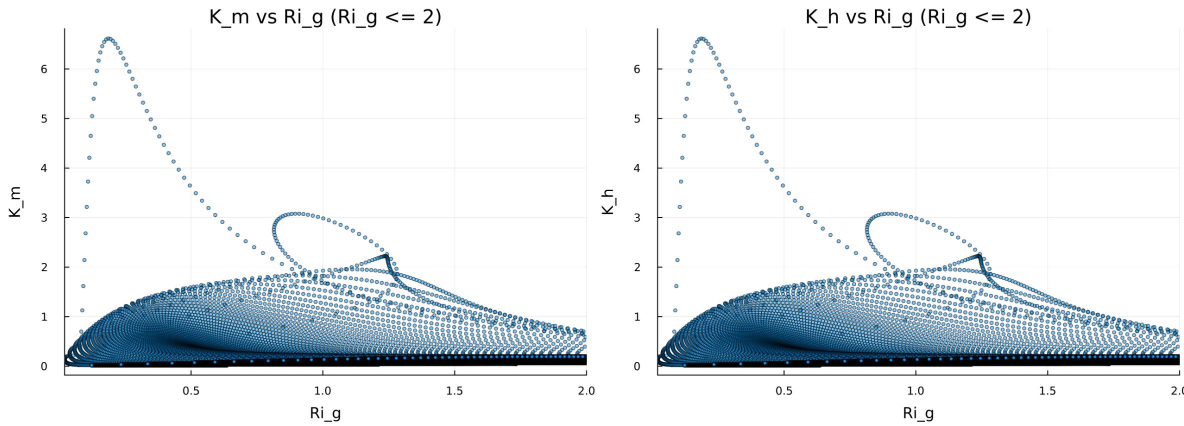

Figure 53: Fold proximity diagnostics versus time

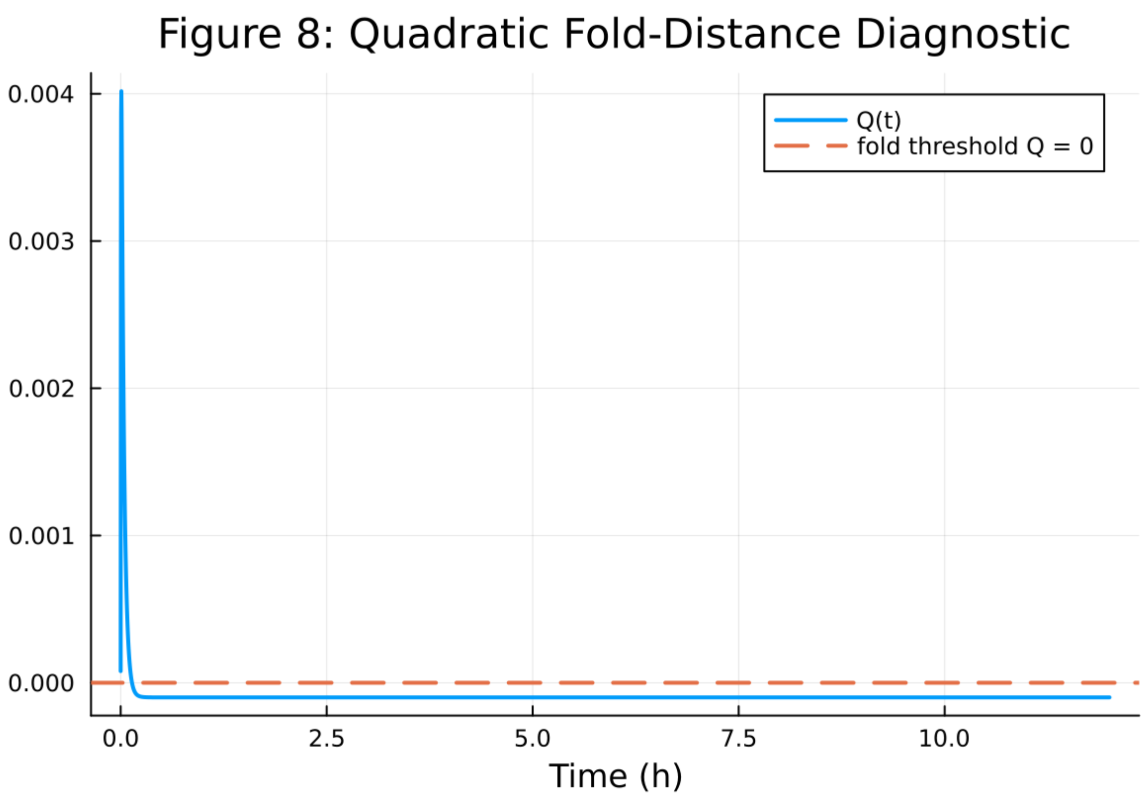

Figure 54: fig08 fold proximity.png

Table 7: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	7.000000 $\mathrm{m\,s^{-1}}$
Meridional geostrophic wind	V_g	0.000000 $\mathrm{m\,s^{-1}}$
Surface roughness (momentum)	z_{0m}	1.000e-04 m
Surface roughness (heat)	z_{0h}	1.000e-05 m
Coriolis parameter	f	1.000e-04 $\mathrm{s^{-1}}$
Reference potential temperature	θ_a	257.000000 K
Deep soil temperature	T_{deep}	255.500000 K
Downward longwave forcing	$R_{\downarrow}$	190.000000 $\mathrm{W\,m^{-2}}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{\mathrm{min,surf}}$	0.001000 $\mathrm{m^2\,s^{-1}}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	257.000000 K
Top relaxation coefficient	λ_{top}	0.000000 $\mathrm{s^{-1}}$

15.2 Forcing and Boundary Conditions

15.3 Numerical Verification

- Maximum surface energy closure error: 0 $\mathrm{W\,m^{-2}}$
- Fold-proximity fraction: 0.393427%
- Diffusivity bounds: $K_m \in [0.018229, 7.434071]$ $\mathrm{m^2\,s^{-1}}$
- Richardson number bounds: $Ri \in [0.020911, 2.973 \times 10^4]$

15.4 Generated Artifacts

- Time-series CSV: `results/sheba_high_top/time_series.csv`
- Diagnostic summary JSON: `results/sheba_high_top/summary.json`
- Payload JLD2: `results/sheba_high_top/payload.jld2`

15.5 Figure Manifest

1. Time series of T_s, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U, theta, and Km at representative times
5. Surface energy budget components (R_n, H, G, storage)
6. Phase portrait on regularized manifold (Delta vs e_xi)
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

15.6 Included Plots

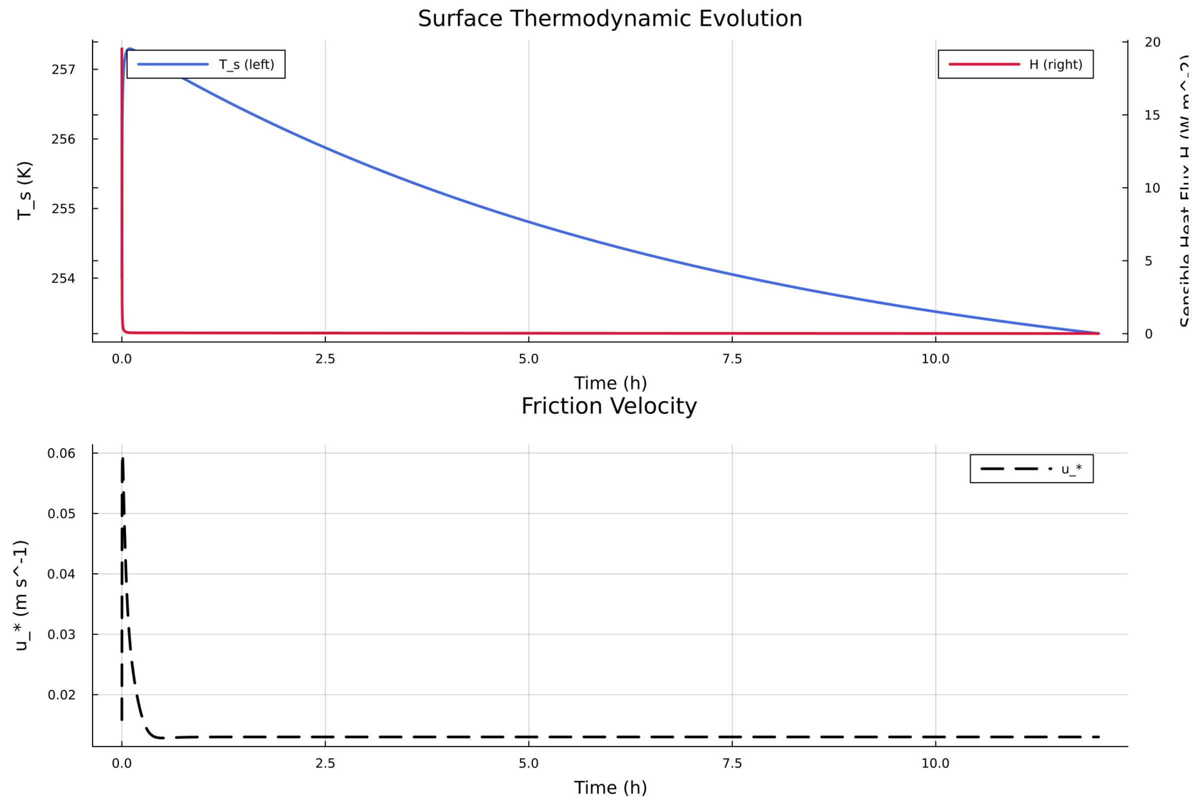

Figure 55: Time series of T__s, sensible heat flux, and friction velocity

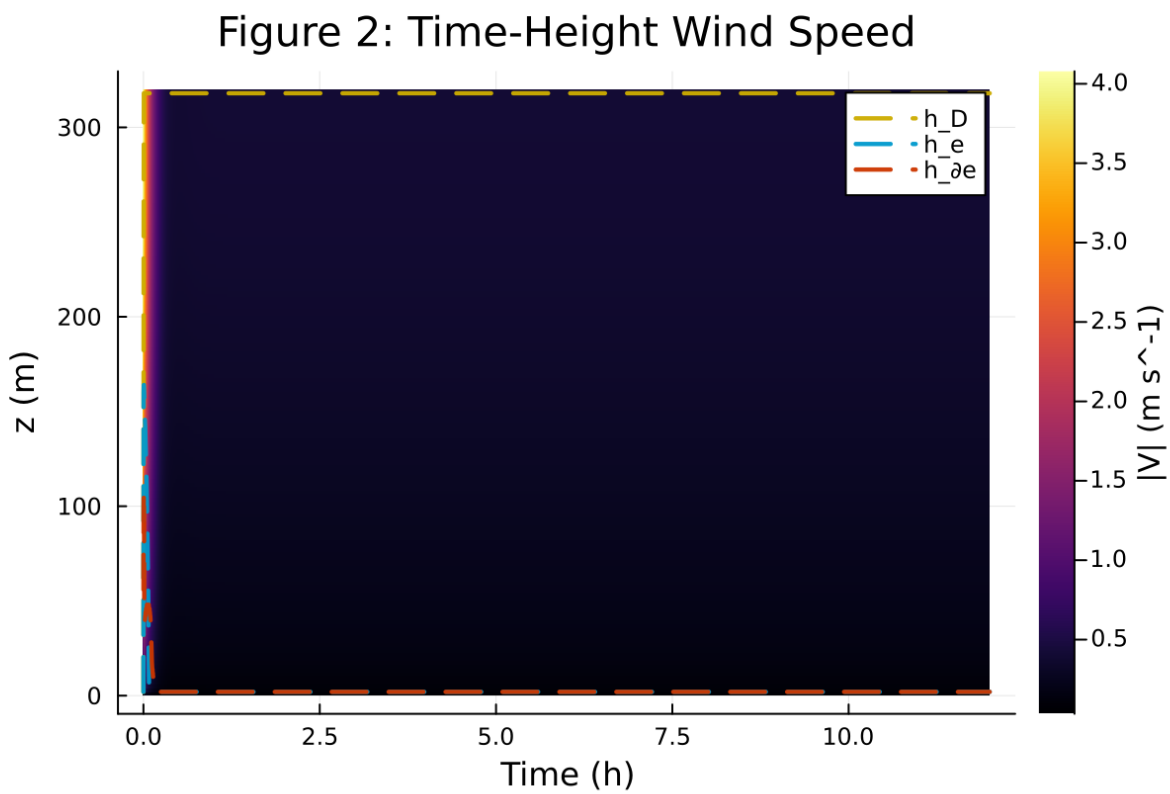

Figure 56: Time-height contour of wind speed with LLJ evolution

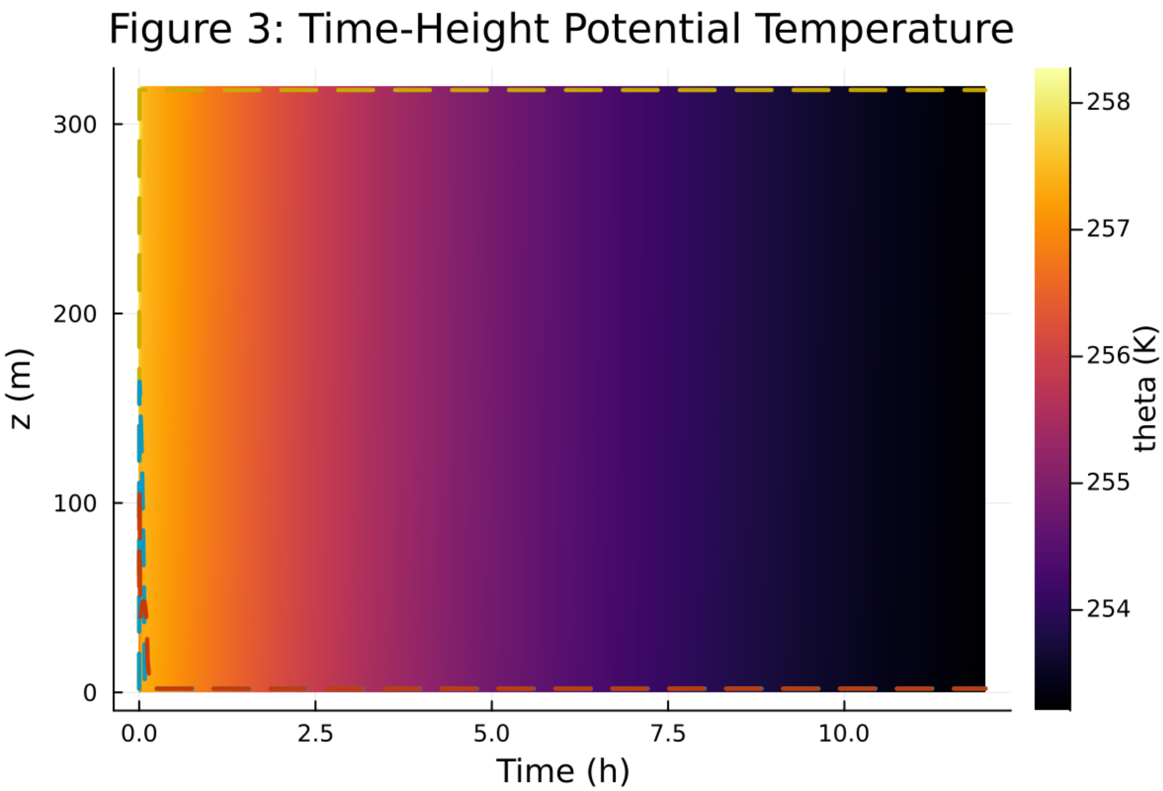

Figure 57: Time-height contour of potential temperature and inversion growth

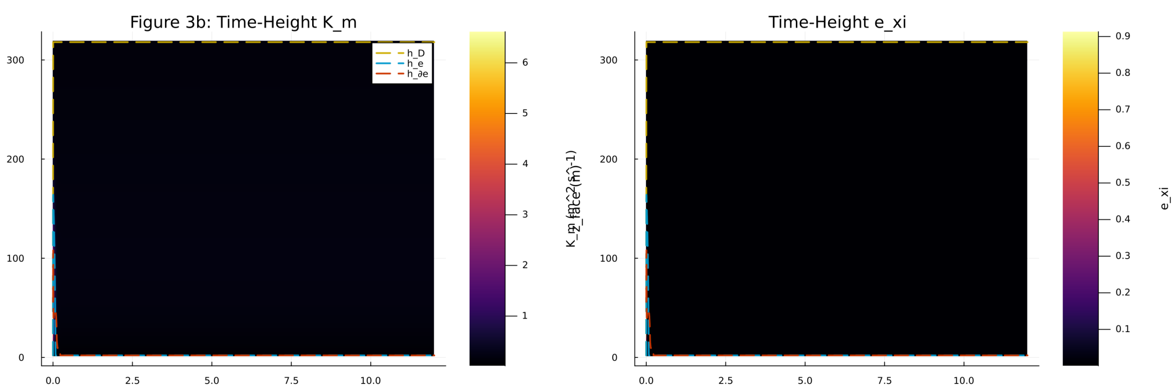

Figure 58: Vertical profiles of U, theta, and Km at representative times

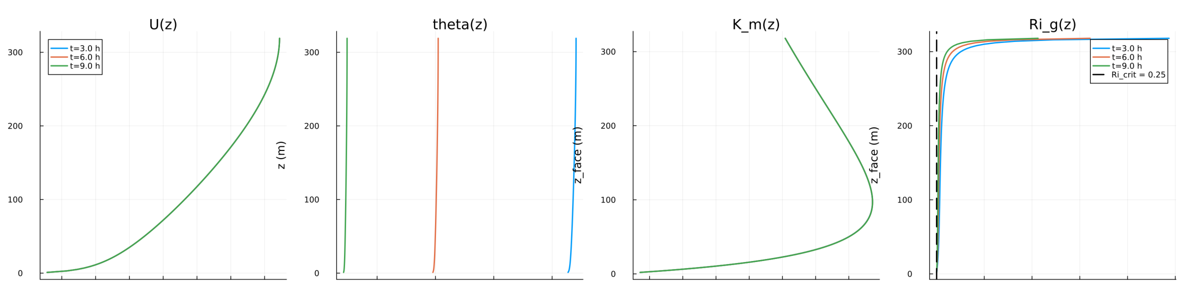

Figure 59: Surface energy budget components (R__n, H, G, storage)

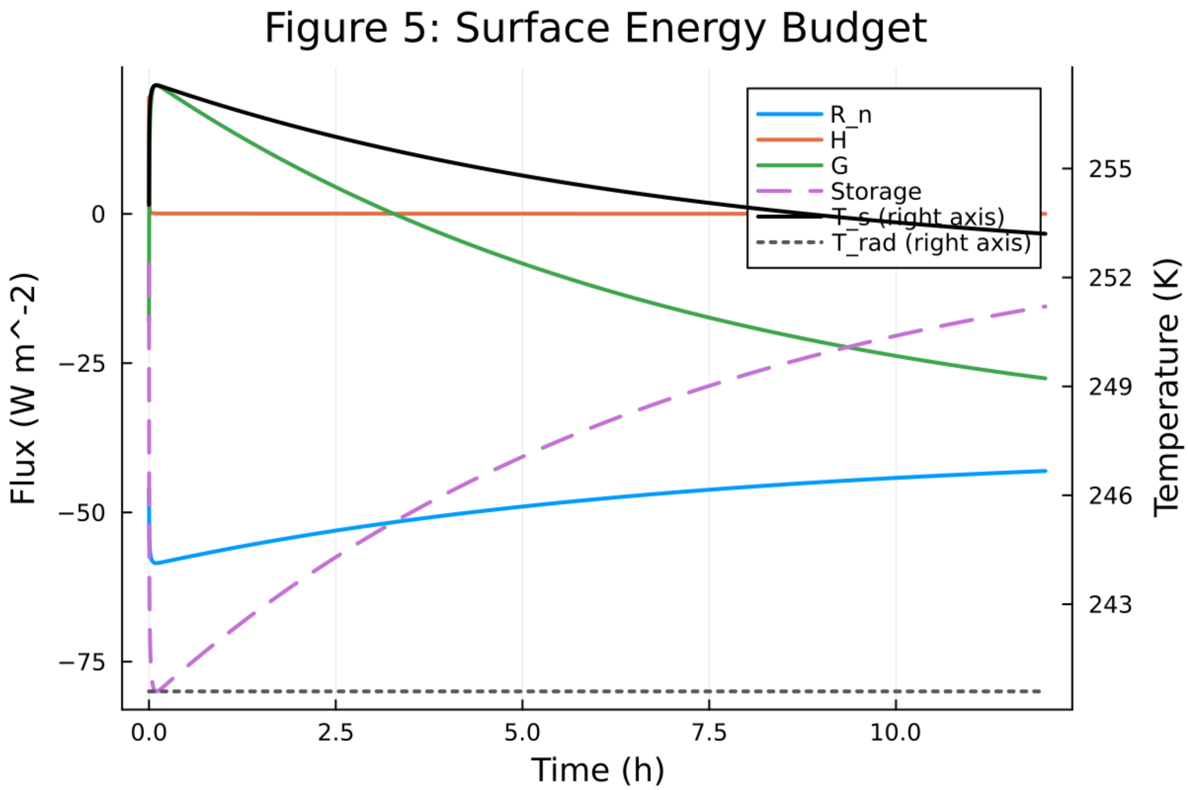

Figure 60: Phase portrait on regularized manifold (Delta vs e__xi)

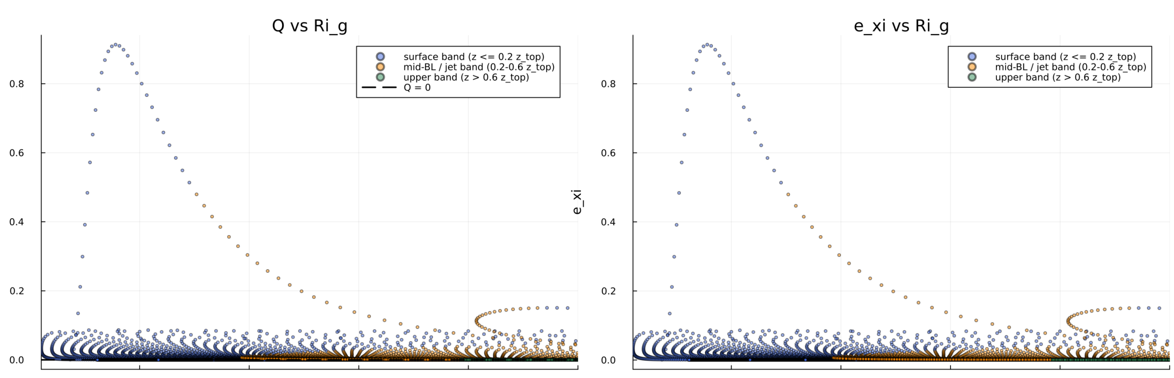

Figure 61: Eddy diffusivity response (Km, Kh) versus stability/Richardson number

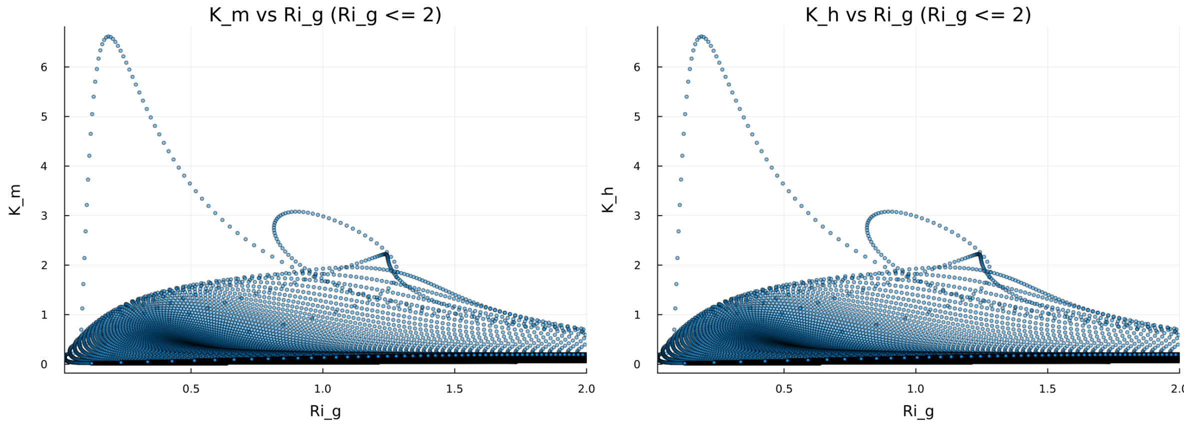

Figure 62: Fold proximity diagnostics versus time

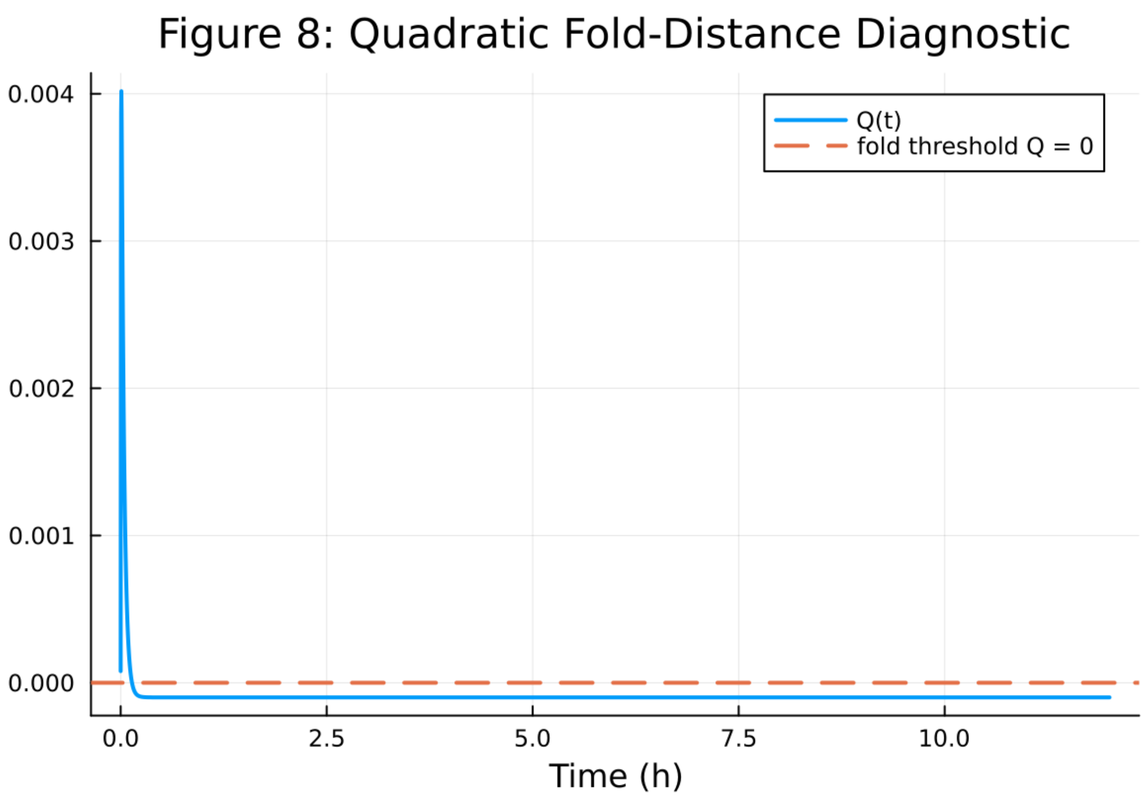

Figure 63: fig08 fold proximity.png